The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Objective. Temporal-interference (TI) stimulation applies two high-frequency currents whose amplitudes beat at a low difference frequency, offering focal, steerable stimulation deep in the brain. Yet an amplitude-modulated field carries no spectral power at the beat frequency, so no passive, linear element of a neuron can follow it. Recovering the beat requires a nonlinearity, which has generally been sought in single-cell ion channels. We ask whether demodulation and its frequency tuning are instead properties of the neural population. **Approach.** We treat the firing-rate nonlinearity of a neural mass, the $\sim 10^4$-neuron unit that generates the EEG, as an amplitude detector, and its recurrent synaptic network poised near a Hopf bifurcation as a resonant amplifier. The account is tested across a modeling ladder: a heuristic Jansen–Rit column (JR), a laminar model (LaNMM), an exact next-generation mean field (NMM2), and the quadratic integrate-and-fire spiking network underlying it. **Main results.** Population transfer function (sigmoid in JR) curvature acts as a square-law detector that recovers the beat, and the network amplifies it resonantly at its own natural frequency. Detection is inherited from the single neuron; the sharp frequency selectivity is emergent, set by proximity to criticality and tunable by connectivity. The mechanism reproduces TI's known behavior: independence of the carrier once membrane polarization is matched, largest response when the beat matches a region's intrinsic rhythm, and, because the resonance amplifies oscillatory timing far more than mean rate, realignment of spike timing without a change in firing rate, as observed in vivo. **Significance.** TI efficacy should be as much a property of the brain as of the device, depending on brain state and regional connectivity. The cortical column behaves as a tuned AM radio receiver, which reframes dose optimization around the target's dynamics and yields falsifiable predictions listed in the paper.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

Contents

1 Introduction

Reaching a deep brain target without stimulating the tissue above it is a long-standing goal of non-invasive neuromodulation, and temporal-interference (TI) stimulation [1] is its most prominent recent proposal. TI applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs (typically $f_1 \sim$ 1–2 kHz, $\Delta f \sim$ 1–100 Hz). Where the two fields overlap, their superposition is an AM signal whose carrier is near f_1 and whose envelope beats at the difference frequency Δf. The appeal is spatial: the envelope amplitude peaks in a focal, steerable region at depth, while superficial tissue sees only the (putatively inert) kHz carrier—a promise of non-invasive, focal, deep stimulation that explains the intense interest.

The difficulty is conceptual rather than technological. Write the field as

$$s(t) = \varepsilon\,[\,1 + m\cos(\Omega t)\,]\cos(\omega_c t), \qquad \omega_c = 2\pi f_c, \quad \Omega = 2\pi\,\Delta f, \tag{1}$$

where m is the modulation depth, Δf the envelope (beat) frequency, f_c the carrier frequency (the mean $(f_1 + f_2)/2$ of the two applied tones, the envelope beating at their difference $\Delta f = f_2 - f_1$), and ε the field amplitude (refined to the post-coupling polarization in §2.2); its spectrum has support *only* at ω_c and $\omega_c \pm \Omega$ (Fig. 3): there is no energy at the envelope frequency Ω. A linear, time-invariant (LTI) system maps a sinusoid to a sinusoid of the same frequency; it can attenuate or phase-shift the three carrier lines but can *never* synthesize a new line at Ω. The membrane's passive RC low-pass—often invoked as "the neuron follows the slow envelope"—therefore cannot by itself demodulate TI. Demodulation requires a nonlinearity. This is the founding lesson of AM radio [2]: a receiver cannot recover the message with linear components alone; it needs a nonlinear element (a square-law device or a diode) to shift spectral energy down to *baseband*—the low-frequency band near 0 Hz where the message lives—followed by a tuned filter to select the station and a low-pass to read the envelope.

Where, then, is the nonlinearity in the brain? The original passive-filtering intuition is now widely regarded as insufficient, since a linear filter cannot create the envelope line [3]. The leading biophysical account places the nonlinearity in single-neuron ion channels: Mirzakhalili et al. [3] show in Hodgkin–Huxley/Frankenhaeuser–Huxley axon models that TI works through nonlinear, ion-channel-mediated rectification, in which asymmetric voltage-gated conductances rectify the carrier and, after membrane low-pass, follow the envelope—the textbook rectify-then-filter detector. Other studies report that TI thresholds inherit the carrier's frequency dependence, suggesting a mechanism shared with direct kHz stimulation (conduction block, onset response) rather than a dedicated envelope channel [4, 5], and in some peripheral preparations the response is not driven by envelope extraction at all [5]. In vivo, TI reliably alters spike *timing* but not firing rate in the primate brain, and is much weaker than conventional low-frequency stimulation under human-relevant conditions [6]. Most tellingly, recent slice work finds that pyramidal and parvalbumin (PV) interneurons respond differently and that most pyramidal TI responses disappear when synaptic transmission is blocked, leading Caldas-Martinez et al. to conclude that TI is "largely a network phenomenon" [7]; computational network models point the same way [8].

Here we show that a population-scale mechanism, built from two ingredients already in standard cortical models, suffices: a static firing-rate *sigmoid*, whose nonzero curvature σ'' makes it a square-law demodulator, and a *second-order-synapse network* whose synaptic kinetics filter the carrier away while a weakly-damped *stable focus* resonantly amplifies the recovered envelope at the network's natural frequency (Fig. 1). The amplification is a property of the focus, not of the Hopf specifically: it sharpens as the focus nears criticality, which the column approaches near *both* the saddle-node on an invariant circle (SNIC, a slower rhythm) and the Hopf (alpha). Where existing accounts locate the nonlinearity in cellular excitability, ours reads it out at the mesoscopic scale: the sigmoid is a coarse-grained summary of the population's single-cell input–output curves, but once coarse-grained it is a *distinct mesoscopic nonlinearity* with enough curvature to

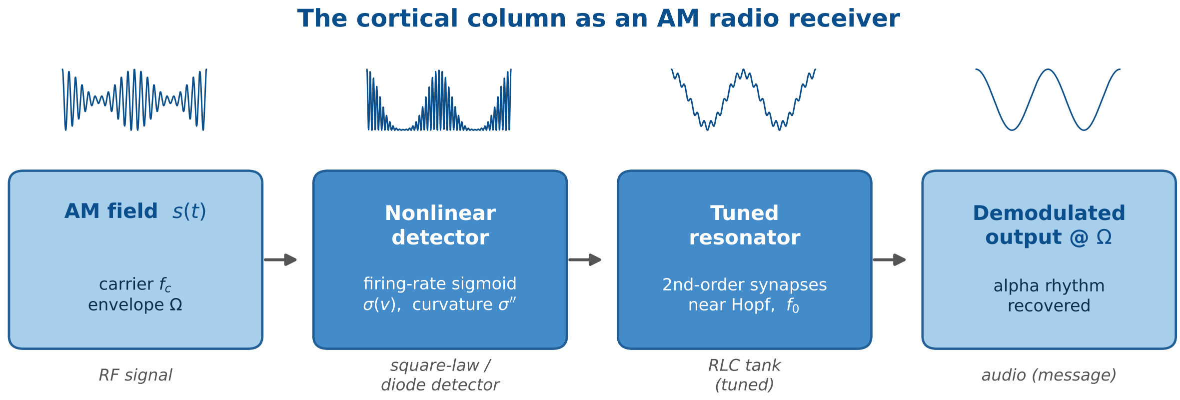

Figure 1: The cortical column as an AM radio receiver (schematic). An AM field (carrier f_c, envelope Ω) reaches a nonlinear detector (the firing-rate sigmoid, curvature σ''), whose output passes through a tuned resonator (the second-order-synapse network near a Hopf, natural frequency f_0), recovering the envelope as an oscillation at Ω. The stages map onto the radio-frequency (RF) front end, square-law/diode detector, resonant RLC tank (the tuned resistor–inductor–capacitor circuit that selects one station), and audio output of a classic AM receiver [2]. The firing-rate sigmoid plays the role of the detector; the second-order-synapse network near a Hopf is the tuned tank that selects and amplifies a particular envelope frequency, so TI "broadcasts" to whichever columns are tuned to its beat frequency.

detect envelopes, whatever microscopic mechanism supplies it. This is a network-level mechanism for TI demodulation—carrier-independent (at fixed polarization) and frequency-selective—complementary to the disinhibitory network effect (PV-interneuron silencing) reported by [7]; both place TI's action in the population rather than the single cell. We develop it first in a single Jansen–Rit column, analyzable in closed form, then in the two-band laminar model (LaNMM), whose own alpha and gamma loops bring it closer to cortex, and finally in an exact next-generation mean field, where the demodulating nonlinearity is derived rather than assumed.

This work builds on a line of research in which neural-mass models implement AM-radio principles *internally*, for the brain's own signals. In our cross-frequency-coupling account of predictive coding [9], the LaNMM realizes a columnar Comparator through sigmoid-enabled signal-to-envelope (SEC) and envelope-to-envelope (EEC) couplings, where error evaluation is literally demodulation—the sign-reversed prediction "un-modulates" the carrier. The Hierarchical Amplitude Modulation (HAM) framework [10] generalizes this to nested envelopes with constant-Q (log-spaced) bands, a candidate origin of $1/f$ spectra—again through the sigmoid's quadratic cross-term. The present work turns this machinery outward: if the cortex is an AM transceiver for its own rhythms, it can also be an AM receiver for an external carrier.

2 Theory

2.1 Neural mass models and the Jansen–Rit circuit

A neural mass model (NMM) collapses the activity of a large population ($\sim 10^4$) of similar neurons into a handful of population-averaged variables [11, 12]. Two elements recur in every such model. First, each synapse acts as a *linear filter* $\hat{K}$ that converts an incoming presynaptic firing rate $r(t)$ into a postsynaptic potential (PSP) $x(t) = \hat{K}[r]$; equivalently $r = \hat{L}[x]$ with $\hat{L} = \hat{K}^{-1}$ a differential operator whose few coefficients set the synapse's gain, decay time and delay [11]. In the Jansen–Rit (JR) family the kernel is the "alpha function" $h(t) = Aa\,t\,e^{-at}$ (excitatory; B, b for the inhibitory synapse), i.e. the second-order operator $\hat{L} = (Aa)^{-1}(\mathrm{d}/\mathrm{d}t + a)^2$, so each PSP obeys $\ddot{x} + 2a\dot{x} + a^2 x = Aa\,r(t)$. Second, the PSPs converging on a population sum to a mean membrane potential $v = \sum_s x_s$, which a static, saturating *transfer function*—the sigmoid—turns back into a population firing rate [13],

$$\sigma(v) = \frac{2e_0}{1 + e^{\rho(v_0 - v)}}, \tag{2}$$

with maximal firing rate $2e_0$, half-maximal at the inflection v_0, and slope set by ρ; it saturates because firing rates are bounded [11]. The sigmoid is the only nonlinearity in the model, and—as we show—it is what makes (de)modulation possible. Its curvature, which does the demodulating, originates in the single neuron's own input–output nonlinearity: σ is the mean-field (ensemble) average of single-neuron transfer curves over a heterogeneous, noisy population, so σ'' is inherited from single cells and only reshaped—smoothed and broadened—by the spread of states, thresholds and noise: the curvature is cellular in origin, but its demodulation is read out at the population level by the generic sigmoid, while the network supplies the resonant *amplification* we use here (§2.3). Frequency-selective resonance can also be intrinsic to single cells (e.g. I_h and I_M currents [14]); we locate it instead in the network, where second-order synapses near criticality generate it even when the cells do not [15, 16].

This becomes exact one rung up the modeling ladder [11]. For a network of all-to-all coupled quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities, the Ott–Antonsen ansatz yields a *closed, exact* mean field for the population firing rate r and mean potential v—the next-generation (Montbrió–Pazó–Roxin, MPR) mass [17], completed with the second-order synapses of neural-mass models in the exact "NMM2" theory [18, 19]:

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t), \tag{3}$$

with the applied current $I(t)$ entering linearly, the common background excitability $\bar{\eta}$ (the bifurcation parameter), and the heterogeneity half-width Δ setting the smoothing. Here the demodulating nonlinearity is *derived, not postulated*: the quadratic v^2 term (with the rv coupling) is an exact square-law rectifier of the input, inherited directly from the single-neuron QIF dynamics. The Jansen–Rit sigmoid is the heuristic (NMM1) stand-in for this transfer—exact in the slow-synapse limit [19]—so our square-law result is a generic property of the mean field, not an artifact of expanding one postulated curve. The same exact theory already exhibits weak-field resonance and Arnold tongues that sharpen with self-coupling [19, 18], the near-Hopf amplification we exploit below.

What the transfer functions represent. In every model the demodulating object is a *population* input–output map rather than a passive somatic membrane element. In JR/LaNMM it is the sigmoid (2), the effective transfer of an excitable neuronal ensemble (mean PSP in, population firing rate out); in NMM2 the same role is played by the exact MPR quadratic (3). Its curvature is the mesoscopic trace of cellular nonlinearities (thresholds, spike generation, refractoriness, noise, saturation), so reading the demodulation off σ'', or off the derived v^2 term, is legitimate at this scale. The membrane low-pass does not contradict this: it bears on *access*, not legitimacy. The open biophysical question is which fast compartment (axon initial segment, node, terminal, or channel) lets a kHz carrier reach that nonlinear transfer (§2.5); once the carrier is sampled there, the neural mass describes the population-level demodulation and near-Hopf amplification of the recovered envelope.

The single-column JR circuit couples a pyramidal population to one excitatory and one inhibitory interneuron population through this synapse+sigmoid motif. Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, and the local-field-potential (LFP) proxy $v = y_1 - y_2$, with external input p,

$$\ddot{y}_0 = Aa\,\sigma(y_1 - y_2 + s(t)) - 2a\dot{y}_0 - a^2 y_0, \tag{4}$$
$$\ddot{y}_1 = Aa\left[\,p + C_2\,\sigma(C_1 y_0)\,\right] - 2a\dot{y}_1 - a^2 y_1, \tag{5}$$
$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{6}$$

Table 1: Jansen–Rit parameters [12].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

integrated as six first-order ODEs with the parameters of Table 1 [12]. The term $s(t)$ is the applied-field perturbation, introduced next; the read-out is $v = y_1 - y_2$.

For the biophysical realization (§4.5) we use a second model built from the same ingredients: the laminar neural mass model (LaNMM) [9, 20], which embeds this synapse+sigmoid motif in a two-band cortical column—a deep Jansen–Rit-like subcircuit (pyramidal P_1 with spiny-stellate (SS) and somatostatin (SST) interneurons; intrinsic alpha, $\sim$10 Hz) coupled to a superficial PING-like (pyramidal–interneuron network gamma) subcircuit (pyramidal P_2 with PV interneurons; intrinsic gamma, $\sim$40 Hz). The single JR column lets us isolate and analyze the mechanism; the LaNMM then exhibits it in a column with its own fast/slow loops and two native resonances. Everything below applies to both: the field couples through a sigmoid, and the recovered envelope drives a resonant circuit.

2.2 From applied field to a demodulated envelope

We model the applied TI field $E(t)$ (V/m) as a quasi-static perturbation of the membrane potential that the spike-generating nonlinearity reads. "Quasi-static" means we neglect electromagnetic propagation and induction (wavelengths $\gg$ head, frequencies far below tissue relaxation), so the field acts as an instantaneous polarization; taking it to add directly inside the sigmoid argument,

$$\varphi(t) = \sigma\big(v(t) + \lambda\, E(t)\big), \qquad E(t) = E_0\,[\,1 + m\cos\Omega t\,]\cos\omega_c t \tag{7}$$

This is the standard weak-field coupling $\delta V_m = \vec{\lambda}\cdot\vec{E}$ of transcranial-stimulation modeling [11], with λ a lumped polarization length. Three quantities must be kept distinct: the applied field $E(t)$, the induced membrane polarization $\delta V_m(t)$ *after* cable/membrane filtering, and the argument of the firing-rate map. The chain is

$$E(t) \xrightarrow[\text{cable/membrane}]{} \delta V_m(t) = \kappa(\mathbf{x},\theta,f_c)\,E(t) \xrightarrow[\text{nonlinearity}]{} \sigma\big(v + \delta V_m\big) \xrightarrow[\text{network}]{} \text{resonance}, \tag{8}$$

with coupling $\kappa(f_c) = \lambda\,|H_m(f_c)|$ (mV per V/m) lumping cell geometry, orientation and the effective polarization length λ with the membrane band-pass H_m (§2.5). We therefore define ε as the *post-coupling polarization* amplitude that actually reaches the nonlinearity, $\varepsilon \equiv \kappa(f_c)E_0 = \lambda|H_m(f_c)|E_0$, with $s(t) \equiv \delta V_m(t)$, recovering Eq. (1)—not the raw applied field. In the main text we take the favorable quasi-static, direct limit $\kappa = \lambda$ ($|H_m|\to 1$), which isolates the sigmoid+resonance principle and is appropriate when a fast nonlinearity (axonal/nodal/terminal) samples the carrier before RC attenuation; §2.5 restores $H_m(f_c)$ and shows that the carrier dependence enters quadratically, $A_\Omega \propto |H_m(f_c)|^2$.

Expanding σ about an operating point v^* (a Volterra/Taylor expansion, as in [10]) gives $\sigma(v^*+s) \approx \sigma(v^*) + \sigma'(v^*)\,s + \frac{1}{2}\sigma''(v^*)\,s^2 + \ldots$ The truncation holds only while $|s|$ is small against the sigmoid's voltage scale $1/\rho \approx 1.8$ mV. Human TI polarizations are $\sim 10^{-2}$ mV, over two orders below that, so the quadratic term is the leading nonlinearity throughout the regime this paper treats. Rodent TI is not in that regime: the fields are $\sim 10^3$ times larger, the sigmoid saturates, and the expansion fails. §5 returns to what that costs when rodent results are scaled to humans. Intuitively, the difficulty is that the field carries power only near the carrier ω_c, far from the slow envelope frequency Ω, so some operation must move power down to Ω. The *linear* term cannot: it passes the input lines straight through, reproducing the carrier and its two flanking *sidebands* (the lines at $\omega_c \pm \Omega$ created by the modulation), but still nothing at Ω. The *quadratic* term can: squaring the field multiplies the fast carrier by itself, and because $\cos^2$ has a constant (zero-frequency) part, the squaring folds power down to *baseband*—the low-frequency band near 0 Hz, where the slow envelope lives. This is *rectification*, exactly what a diode does in an AM radio. Keeping the baseband (low-frequency) part of $s^2 = \varepsilon^2(1 + m\cos\Omega t)^2 \cos^2\omega_c t$ (the fast $\cos^2\omega_c t$ averages to its mean $\frac{1}{2}$),

$$\tfrac{1}{2}\sigma''\, s^2\big|_{\mathrm{LF}} = \tfrac{1}{4}\sigma''\varepsilon^2\Big[\big(1 + \tfrac{m^2}{2}\big) + 2m\cos\Omega t + \tfrac{m^2}{2}\cos 2\Omega t\Big], \tag{9}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon^2\, m. \tag{10}$$

Physical TI applies two tones rather than the modulated carrier of Eq. (7). The two waveforms are not the same signal: the two-tone sum is suppressed-carrier, with lines only at ω_1 and ω_2, whereas Eq. (7) also carries a line at ω_c. What they share is their demodulated output. Writing the post-coupling polarizations as ε_1 and ε_2, the same expansion gives $A_\Omega = \frac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2$: the effective amplitude is the geometric mean $\sqrt{\varepsilon_1\varepsilon_2}$, and at fixed total the drive peaks where the two beams are matched, which is where TI's focality comes from. Since A_Ω agrees between the two forms at matched peak polarization, we use the modulated-carrier form throughout.

In the language of [10], the quadratic cross-term $\kappa_2\, s_1 s_2$ between a slow and a fast input implements amplitude modulation/demodulation and the associated intermodulation lines. Demodulation is thus itself an intermodulation product: the line at Ω is the *difference frequency* of the two applied carriers (ω_c and $\omega_c + \Omega$), generated by the same quadratic term that—acting a second time, now on an intrinsic rhythm ω_0—produces the $\omega_0 \pm \Omega$ sidebands seen at the spiking level (Fig. S15b). One nonlinear mixer, applied in cascade. Equation (10) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \kappa(f_c)E_0$, so the field required for a given effect rises with f_c (§2.5)—consistent with the experimental consensus. Carrier independence is thus a statement about the recovered envelope at fixed polarization, not about the applied field.

2.3 Resonance, carrier independence, and the synaptic band-pass

Equation (10) contains no ω_c: the demodulated amplitude is carrier-independent at fixed polarization (§2.2). Under the direct coupling (7) the carrier need only be high enough that the synaptic low-pass suppresses it relative to the envelope, and "inert"—it must not itself fall in, or beat into, the network's responsive band, nor engage other carrier-specific mechanisms (the ion-channel/kHz pathway). Within those bounds the response is flat in f_c (Fig. 5, left). The synaptic suppression is concrete: each JR synapse applies the operator $\hat{L}_s$ with impulse response $h(t) = Aa\,t\,e^{-at}$, i.e. transfer function $H(\omega) = Aa/(i\omega + a)^2$, a critically-damped second-order low-pass with corner $a/2\pi \approx 16$ Hz. It passes the $\sim$10 Hz envelope ($|H| \sim 0.7$) while suppressing a 100 Hz carrier to $\sim$2.5% (Fig. 5, right), so the carrier is filtered out of the loop *after* the sigmoid has used it to demodulate. Thus, two distinct mechanisms shape the closed-loop dynamics: the sigmoid extracts an envelope whose amplitude is inherently independent of the carrier, and the synaptic low-pass subsequently removes the carrier itself from the spectrum.

What turns this recovered envelope into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives a leading eigenvalue pair $\lambda_\pm = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, so the whole network behaves as a single weakly damped harmonic oscillator with damping γ and natural frequency ω_0 (the gain below is its one-mode, center-manifold reduction). Just below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated envelope (10) at Ω gives the textbook resonance gain

$$G(\Omega) \propto \left[(\omega_0^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0}, \tag{11}$$

so the steady response $\sim A_\Omega\, G(\Omega)$ is a Lorentzian peaked at $\Omega = \omega_0$ whose height diverges as the Hopf is approached. Just above the Hopf the mass is an autonomous limit cycle at ω_0, and the demodulated drive *entrains* it (an Arnold tongue), giving a response peak near ω_0 that grows with cycle amplitude.

2.4 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 6). The Population Excitation–Inhibition Index (PEIX) of [9] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\mathrm{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \operatorname{sech}^2\!\left[\tfrac{\rho}{2}\,(v^* - v_0)\right] \in (0, 1], \tag{12}$$

a dimensionless read-out of where v^* sits, the side given by $\operatorname{sign}(v^* - v_0)$. PEIX thus tracks σ' (maximal at the inflection v_0), whereas the demodulation gain tracks σ'' (zero at v_0, peaked on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks exactly where the demodulation gain vanishes. PEIX is therefore a convenient, measurable read-out of the operating point and hence an indirect predictor of σ'' and of TI efficacy. The mechanism predicts that anything shifting v^*—neuromodulation, attention, arousal, or pathology (psychedelics, PV interneuron dysfunction in Alzheimer's disease)—should scale, and can even flip the sign of, the TI response: a falsifiable handle.

2.5 Biophysical realization: carrier frequency, fast elements, and state

Carrier independence (at fixed polarization) holds for the direct coupling (7). Real somata are not direct: the membrane (and dendritic cable) low-passes the field before the spike nonlinearity. (The demodulating σ'' is the *population* transfer; the membrane filter is the upstream *access* stage—this section asks only whether the carrier survives to it.) Modeling this as a first-order filter $H_m(\omega) = 1/(1 + i\omega\tau_m)$, the carrier amplitude reaching the detector is $\varepsilon_{\text{eff}}(f_c) = \varepsilon/\sqrt{1 + (\omega_c\tau_m)^2}$, and since $\Omega \ll \omega_c$ the three AM lines are attenuated almost equally, so

$$A_\Omega(f_c) \;\simeq\; \tfrac{1}{2}\,\sigma''(v^*)\, m\, \varepsilon_{\text{eff}}^2 \;=\; \frac{A_\Omega(0)}{1 + (\omega_c\tau_m)^2} \;\xrightarrow[\omega_c\tau_m \gg 1]{}\; A_\Omega(0)\,\frac{1}{(\omega_c\tau_m)^2}. \tag{13}$$

The demodulated response rolls off as $1/f_c^2$ above the membrane corner (Fig. S7). For a pyramidal soma/dendrite the passive time constant is $\tau_m \sim 10$–30 ms (human L2/3 pyramidal cells: $\sim$12 ms [21] and 16.5 ± 3.7 ms [22], which bracket the $\sim$16 ms we adopt for the soma, corner $\sim$10 Hz), so the carrier-power factor $|H_m|^2$ is only $\sim 10^{-4}$ at 1 kHz, $\sim 2.5 \times 10^{-5}$ at 2 kHz and $\sim 10^{-6}$ at 10 kHz. Passive somatic demodulation is therefore negligible at TI's kHz carriers—a four- to six-order-of-magnitude suppression.

The escape is that the demodulating nonlinearity need not sit at the soma. At a fast element (axon, axon initial segment (AIS), node of Ranvier; $\tau \approx 0.2$ ms, effective corner $\sim$800 Hz) the carrier is only mildly attenuated in the kHz band ($|H|^2 \approx 0.39$ at 1 kHz, 0.14 at 2 kHz, still $\sim 6 \times 10^{-3}$ at 10 kHz; Fig. S7). A square-law nonlinearity there produces a real envelope-frequency current, and that low-frequency component is *not* attenuated by the membrane—it is exactly what the slow network reads. The per-neuron modulation is small, but two factors make it count: the envelope sits in the band the membrane passes, and the population near a Hopf amplifies it by $G \sim 1/\gamma$ (Eq. (11)). Demodulation at fast elements, resonantly amplified at the population, is thus a route for TI at realistic carriers—consistent with ion-channel rectification at the node [3] but adding network resonance; our direct-coupling demonstration is the $\tau \to 0$ idealization of this limit, and a direct kHz-carrier run (Fig. S12) confirms it.

Because transcranial electrical stimulation (tES)—and TI, a tES with a kHz carrier—is weak and *modulatory*, the fast-element route is a small, sustained bias of the most excitable compartments, not "spiking the AIS at the beat frequency." Since the applied field adds to endogenous drive, a weak bias near threshold shifts spike probability and timing through stochastic resonance [23, 24]. The population sigmoid (2) is itself a mean-field stochastic-resonance device—its smooth slope is the spread of thresholds and noise across the population—so the bias shifts the mean rate through σ' while the demodulated envelope emerges through the curvature σ''.

Both halves are state-dependent—detection through $\sigma''(v^*)$, amplification through $1/\gamma$—so TI and transcranial magnetic stimulation (TMS) efficacy should track brain state, not the stimulus alone. The 100 Hz carrier used throughout is not merely convenient: the membrane attenuation is mild even for moderately fast elements ($|H_m|^2 \approx 0.98$ at $\tau = 0.2$ ms, $\sim$0.1 at $\tau = 5$ ms), so a protocol with $f_c \sim 100$ Hz and Δf swept through a region's band is a clean near-term test. The trade-off is spatial: a 100 Hz carrier is less inert off-target than a kHz one (whose subthreshold carrier is the source of TI's focality), so 100 Hz probes the mechanism while kHz optimizes selectivity.

3 Methods

3.1 Single-column Jansen–Rit: integration, lock-in, and controls

(All symbols and units are collected in App. A, Table 3.) We integrate the single-column JR model of §2.1 (Eqs. (4), parameters in Table 1) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (4), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at Ω in v is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). For each measurement the system is settled for 10–16 s (longer near the Hopf, where damping is weak) before a 3–20 s measurement window; the resonance sweeps use a 20 s window so the lock-in averages the limit-cycle beating on the entrainment side. Parameter sweeps are vectorized: the state is an $(N, 6)$ array advanced simultaneously for all N parameter combinations, so a full resonance map runs in seconds. The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically and diagonalizing it: the Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\lambda_+$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \mathrm{Im}\,\lambda_+/2\pi \approx 11.1$ Hz—consistent with the continuation and with the JR bifurcation analysis of [25]. For $p > 315$ the fixed point is a stable focus (resonator); between the SNIC ($p \approx 114$) and the Hopf the column free-runs on the alpha limit cycle.

The response at the envelope frequency is read out with a software lock-in amplifier referenced to Ω—a standard way to isolate the component of a signal at one chosen frequency: multiply by sine and cosine at Ω and average, so everything *not* at Ω cancels and only the envelope-locked amplitude survives. Concretely we multiply $v(t)$ by quadrature references and integrate over a

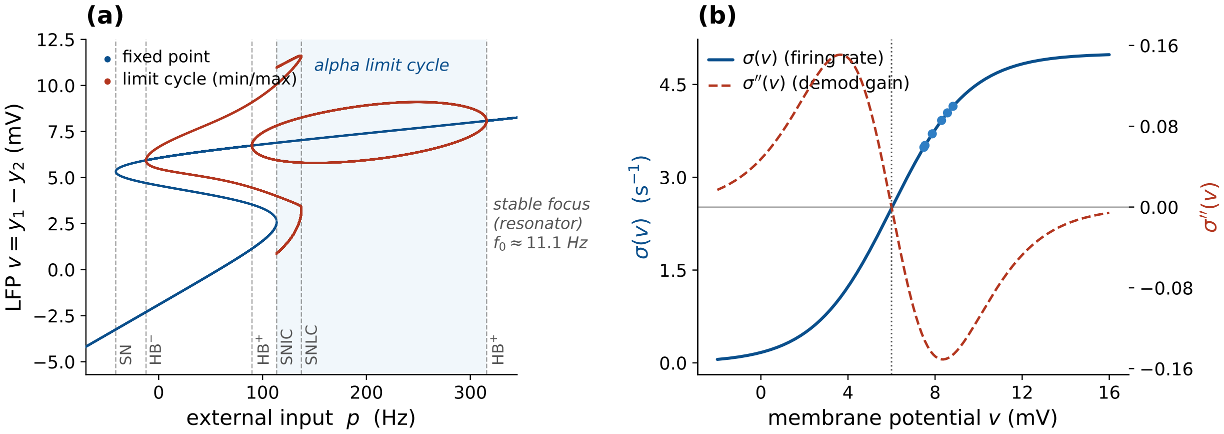

Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature. *(a)* JR bifurcation diagram from numerical continuation—the LFP $v = y_1 - y_2$ along the fixed-point branch (blue) and the min/max of the limit-cycle branches (red) vs. external input p, with *solid* stable and *faded* unstable. The alpha limit cycle (shaded) is bounded by a saddle-node-on-invariant-circle (SNIC, $p \approx 114$) and the supercritical Hopf ($p \approx 315$, $f_0 \approx 11.1$ Hz); beyond the Hopf the fixed point is a stable focus—the near-critical resonator regime used throughout. The column is a weakly-damped focus near *both* bifurcations, so it resonates on either approach—more slowly near the SNIC. Further codim-1 points are marked (saddle-node SN, Hopf HB$^{\pm}$, saddle-node of limit cycles SNLC). *(b)* The sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection v_0. Dots mark the operating points reached by the closed-loop model for the p values used here.

Hann-windowed measurement window to obtain the in-phase (I) and quadrature (Q) components,

$$I = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k \, (v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2}, \quad (14)$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to -A\sin\phi$, so $A_\Omega \to A$ exactly—the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. The one subtlety is that out-of-band components must not leak into the estimate; the dangerous one is the DC level of v ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). Over an exact integer number of Ω-periods $\sum_k w_k \cos \Omega t_k = 0$ and there is no leakage, but in general it is nonzero, so we subtract $\bar{v}$ explicitly; this makes the metric exact for a tone and robust down to the 10^{-8} level needed for the kHz analysis (§2.5). The signed in-phase component I alone recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 6).

Three auxiliary constructions support the main results. Throughout we contrast the *open-loop detector*—the firing-rate sigmoid alone at a fixed operating point v^*, with no recurrence—against the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the AM field through σ at v^* and applies Eq. (14), isolating the pure demodulation law (Figs. 5, 6, S7). The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in Eq. (4) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against finite differences. All code is openly available (see *Code and data availability*).

3.2 Laminar two-band model (LaNMM)

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [9, 20], driving a single population with an AM input

$$e(t) = I_0 + A\,[\,1 + \cos\Omega t\,]\cos(\omega_c t + \phi), \tag{15}$$

the LaNMM counterpart of Eq. (1) (modulation amplitude A, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$), with the remaining drives held flat (Fig. S2). Integration uses an adaptive RK45 scheme (2 ms record step). In place of the lock-in (Eq. (14)) we read the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials from the Hilbert envelope over a post-transient window—a phase-robust measure suited to the entrainment regime, where lock-in phase slips would otherwise complicate the read-out. Two sweeps probe the column: an amplitude–beat sweep over $(A, \Delta f)$ at fixed carrier (the Arnold tongue), and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed A, built on the LaNMM module of [9]. For the alpha resonance of Figs. S9, S3 we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (14)) of v_{P_1} at Δf.

3.3 Exact next-generation mean field (NMM2 PING)

We test the mechanism in the exact next-generation mean field (NMM2, §2.1), following the Rosetta Stone formulation of [11] (which places the Montbrió–Pazó–Roxin reduction [17] on a common footing with the JR and LaNMM models used here), where the firing-rate nonlinearity is the derived MPR dynamics of Eq. (3) rather than a postulated sigmoid. We use a two-population excitatory–inhibitory (PING) motif [18, 19, 11]: each population $\alpha \in \{E, I\}$ is an MPR mean field (firing rate r_α, mean potential v_α) with membrane time τ_α, coupled through second-order synapses,

$$\begin{aligned}
\tau_E \dot r_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, \qquad & \tau_E \dot v_E &= v_E^2 + \bar\eta - (\pi\tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + \delta V(t),\\
\tau_I \dot r_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, \qquad & \tau_I \dot v_I &= v_I^2 + \bar\eta - (\pi\tau_I r_I)^2 + \tau_I C\,(s_E - 2 s_I),
\end{aligned} \tag{16}$$

with second-order synaptic activations $s_E = \hat K_a[r_E]$ (excitatory, time constant τ_a) and $s_I = \hat K_g[r_I]$ (inhibitory, τ_g), each obeying the same critically-damped operator as the JR synapse, $\tau^2\ddot s + 2\tau\dot s + s = r$; here C is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows—i.e. the E membrane sees $C\,(s_E - s_I)$ and the I membrane $C\,(s_E - 2s_I)$, the only asymmetry being the doubled $I \to I$ weight $A_{ii} = 2$. The common background drive $\bar\eta$—the same in both populations, with no separate input current—is the bifurcation parameter: raising it past a supercritical Hopf at $\bar\eta \approx 1$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory membrane current as $\delta V(t) = \lambda E(t)$ with the AM form of Eq. (1) (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band); the polarization is expressed in the dimensionless MPR units of v_E, so the carrier is rectified by the exact quadratic v_E^2 term, with no postulated sigmoid. We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (14)) of r_E at Δf over the $(\Delta f, \bar\eta)$ plane. Parameters (Rosetta Stone PING [11]): $\Delta = 1$, global coupling $C = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar\eta$ swept through the supercritical Hopf at $\bar\eta \approx 1$ ($f_0 \approx 55$ Hz).

Two parameters reach this gamma Hopf, and we use both as criticality knobs—just as the external drive p and the coupling C both reach the JR alpha Hopf. The background drive $\bar\eta$ (above) is swept for the resonance map (Fig. S11); the global coupling $C\,(\equiv J)$ is swept for the J-curve (Fig. S13): we vary J at fixed $\bar\eta$ toward the gamma Hopf J^* (located by bisection on the leading Jacobian eigenvalue) and read the lock-in peak of r_E over Δf. Unlike the JR J-curve, *no operating-point pinning is needed*, because here the rectifier is the exact v_E^2 term, whose quadratic coefficient is fixed by construction; sweeping J therefore changes only the damping γ, isolating the amplification cleanly.

3.4 QIF spiking network (microscale of NMM2)

NMM2 is the exact mean field of an all-to-all network of QIF neurons [17, 11, 19], so we integrate that network directly to test the prediction at the spike level (§4.9): $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (16)), excitabilities $\eta_j \sim$ Lorentzian$(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical population rates, and the AM field entering the excitatory membranes. The bifurcation parameter is the common background drive $\bar{\eta}$, with a gamma Hopf at $\bar{\eta}_{\text{Hopf}} \approx 1.1$ (autonomous $f_0 \approx 53$ Hz; these finite-N values sit just above the exact mean-field Hopf $\bar{\eta} \approx 1$, $f_0 \approx 55$ Hz, the small offset being a finite-size effect); the carrier is $f_c = 300$ Hz and the field amplitude $A_f = 35$. Because this is a *gamma* circuit, the TI beat Δf is set in the gamma band: $\Delta f = 55$ Hz *near* the forced gamma resonance ($\bar{\eta} = 0.85$, below the Hopf) and $\Delta f = 42$ Hz in the oscillatory regime ($\bar{\eta} = 1.5$, above the Hopf)—a detuning of ≈11 Hz below the autonomous $f_0 \approx 53$ Hz that places the beat outside the 1:1 Arnold tongue, so the above-Hopf response is gating, not entrainment (§C). Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega/\langle r_E \rangle$ (a stable measure—the off→on lock-in ratio would divide by the near-zero field-free baseline).

3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The coupling sweeps of Figs. 9 and S14, and the transcranial alternating-current stimulation (tACS) control of Fig. S4, use the single JR column with the operating point v^* pinned: for each global connectivity C we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds v^* (hence $\sigma''(v^*)$) fixed, vary $C\ (= J)$ toward the alpha Hopf, and read γ, ω_0 from the 6×6 Jacobian. The field is the AM drive of Eq. (1) for TI (Fig. 9; AC lock-in vs DC mean rate for Fig. S14) or a direct in-band sinusoid $s(t) = \varepsilon \cos \Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S4), each read out by the lock-in of v at Δf.

3.6 Vocabulary: forced response, entrainment, and gating

"Entrainment" is used in several senses, so we fix three terms. *Forced response* is the driven, non-autonomous case below the Hopf, quantified by the lock-in amplitude at the beat Δf (Eq. (14)). *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator, tested by the 1:1 criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency [26]. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [27, 28]. The distinction is load-bearing, since capture and gating are dynamically different and the same word would otherwise cover both; Appendix C gives the metrics and the Arnold-tongue protocol.

4 Results

We present the evidence on three rungs of the modeling ladder: the single Jansen–Rit column (closed form, analyzable), the laminar LaNMM (two native resonances and genuine carrier dependence), and the exact next-generation mean field (NMM2, the demodulating nonlinearity derived rather than postulated). Each rung shows the same pair—square-law demodulation followed by near-Hopf amplification. Two mesoscale signatures then follow that have no single-cell analog: the network's own coupling sets the gain at fixed detection (the J-curve, §4.7), and the resonance amplifies oscillatory *timing* far more than mean *rate* (§4.8)—the spiking-level prediction we test in the QIF network.

4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

The system is oriented by Fig. 2: a clean supercritical Hopf at $p \approx 315$ with $f_0 \approx 11.1$ Hz, and a sigmoid whose curvature peaks well off the inflection, so the closed-loop operating points sit where σ'' is large. The core result is Fig. 3: an input with no spectral power in the alpha band (lines only at f_c and $f_c \pm \Delta f$) produces a clean ∼11 Hz output oscillation locked to the envelope, with a strong alpha peak in the output spectrum absent from the input.

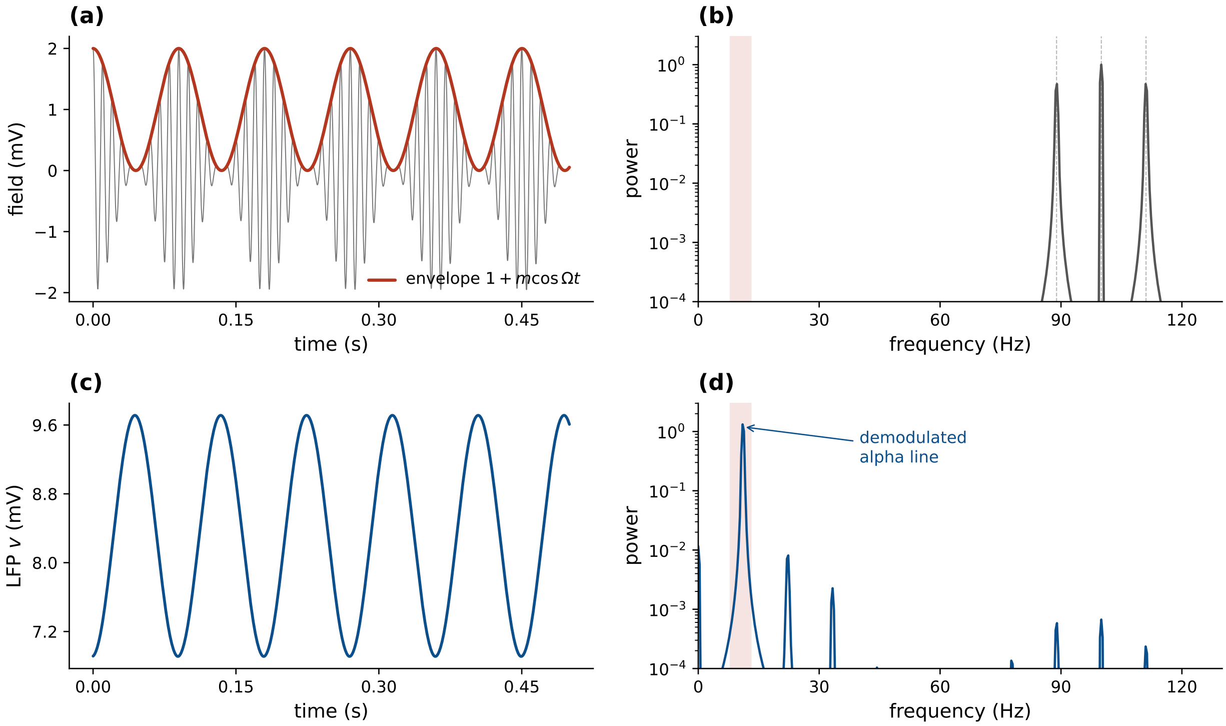

Figure 3: An alpha rhythm is synthesized from a carrier-only input (Jansen–Rit below Hopf). *(a)* The AM field $s(t)$ and its envelope. *(b)* The input spectrum has lines only at f_c and $f_c \pm \Delta f$ (89, 100, 111 Hz); the alpha band (shaded) is empty. *(c)* The output $v = y_1 - y_2$ oscillates at the envelope frequency. *(d)* The output spectrum has a strong peak at $\Omega \approx 11$ Hz, synthesized by the sigmoid and absent from the input.

4.2 Near the Hopf, the demodulated envelope drives a sharp resonance

The resonance is quantified in Figs. 4 and S6. In the fixed-point regime ($p > 315$, a stable focus) the response is a Lorentzian centered at f_0 whose height grows monotonically as the Hopf is approached ($0.23 \to 0.43 \to 0.70$ mV; Fig. 4), i.e. gain $\propto 1/\gamma$. This is genuine forced gain: with no drive the focus is silent at Ω, so all power at Ω is synthesized by the demodulating nonlinearity and amplified by the recurrent loop. The two-dimensional map shows the same story as a ridge at f_0 that sharpens and intensifies toward the bifurcation (Fig. S6). Above the Hopf ($p < 315$) the column free-runs at f_0 and the demodulated drive instead phase-locks that autonomous alpha within a 1:1 Arnold tongue; because lock-in amplitude at Ω then conflates the driven response with the free-running cycle, we characterize that regime with the output frequency f_{out} rather than the lock-in amplitude (Fig. S5).

4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 5). As a function of the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner (Fig. S7): passive somatic demodulation is suppressed four to six orders of magnitude at kHz, whereas a fast element ($\tau \approx 0.2$ ms) retains an appreciable response into the kHz band.

To pre-empt the objection that the demonstration runs only at 100 Hz, we drive the full recurrent column with a genuine kHz-carrier AM field, passing the field through an explicit membrane low-pass (an extra first-order state, time constant τ) for each coupling route before it reaches the sigmoid (Fig. S12a). The column demodulates and resonates at the recovered alpha well into the kHz band when the nonlinearity sits at a fast element ($\tau \approx 0.2$ ms), exactly as Eq. (13) predicts (Fig. S12a). A 2 kHz carrier through the fast element yields a clean alpha line synthesized by the

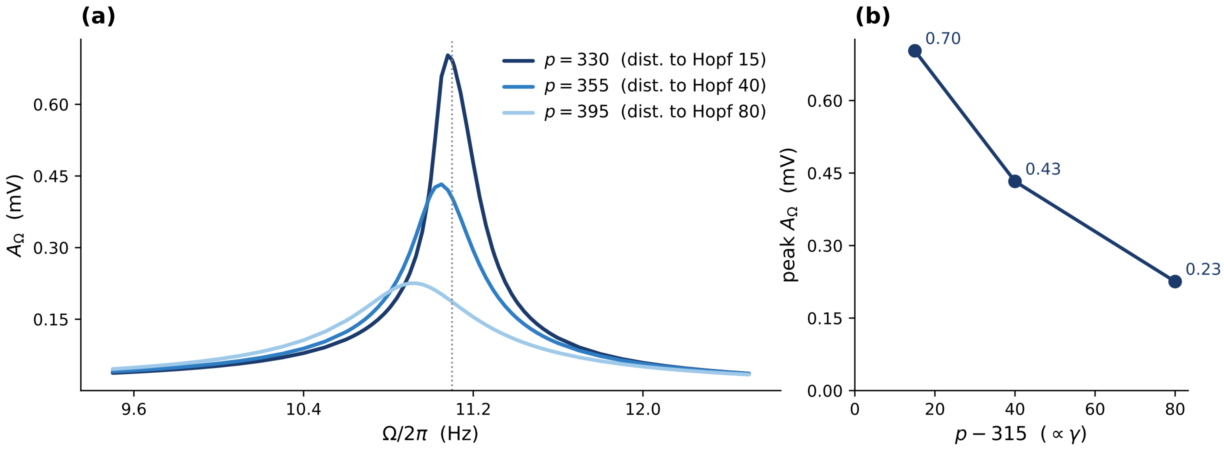

Figure 4: The recovered envelope drives a resonance that sharpens toward the Hopf (Jansen–Rit). Fixed-point regime ($p > 315$, stable focus): lock-in response at Ω vs. envelope frequency ($\varepsilon = 0.3$ mV). *(a)* The demodulated drive forces a Lorentzian resonance at $f_0 \approx 11.1$ Hz whose peak redshifts slightly as the damping grows ($\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\gamma^2}$; $11.08 \to 10.93$ Hz across the three curves); the response carries no power at Ω in the absence of drive, so this is genuine forced gain. *(b)* The peak height grows as the Hopf is approached ($p \to 315$, distance to Hopf $\to 0$). Three points cannot establish a power law; the $1/\gamma$ scaling is quantified over a decade, together with its saturation, in Fig. 9(c). The complementary above-Hopf regime, where the demodulated drive *entrains* the autonomous alpha within a 1:1 Arnold tongue, is characterized with the appropriate observable (output frequency f_{out}, not lock-in amplitude) in Fig. S5.

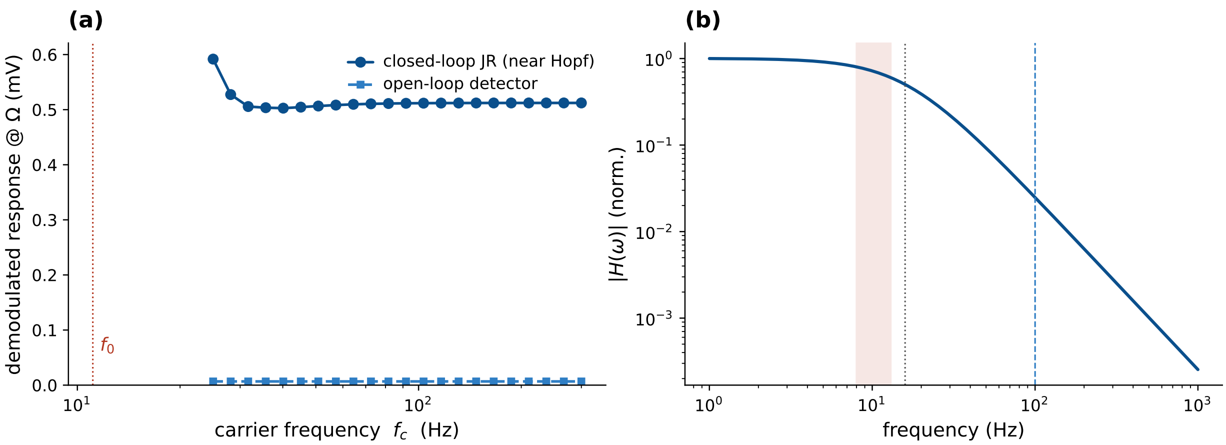

Figure 5: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is flat across carrier frequency $f_c = 25$–300 Hz, both for the open-loop detector (exactly, by Eq. (10)) and for the closed-loop JR near the Hopf (which adds $\sim 60\times$ resonant gain); deviations appear only as f_c approaches the natural band. *(b)* The second-order-synapse band-pass $|H(\omega)|$ passes the envelope (~ 0.7) and suppresses the 100 Hz carrier ($\sim 2.5\%$). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

column from an input with zero alpha power (Fig. S12b). That the bare unmodulated kHz carrier is itself weakly effective—the premise of TI's focality—is consistent with the limited sensitivity of hippocampal network oscillations to unmodulated kHz fields [29].

4.4 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 6)—demodulation is the sigmoid-curvature term of Eq. (10), and the closed-loop operating points sit where σ'' is large and negative.

Two controls close the argument (Fig. S8): the response scales as ε^2 at small drive (measured log–

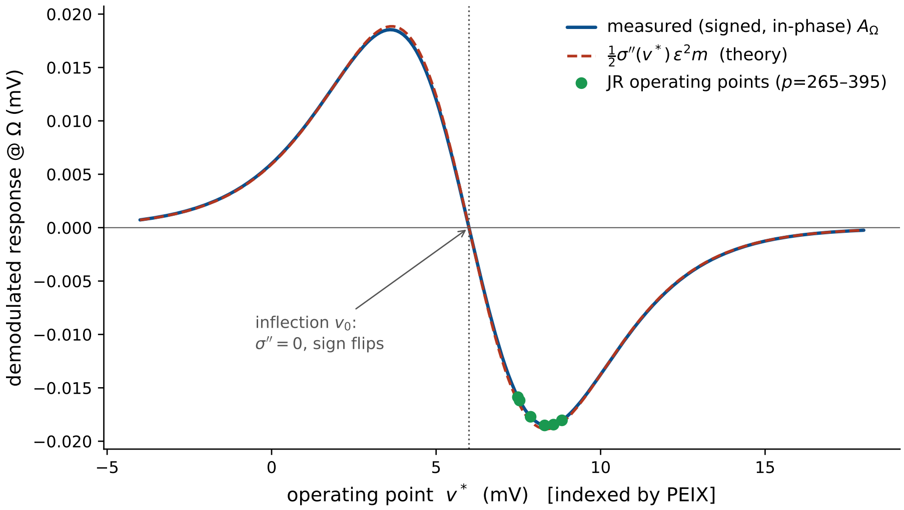

Figure 6: Operating-point law and the σ'' control (Jansen–Rit). The measured signed (in-phase) open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the whole range, vanishes at the inflection v_0, and reverses sign across it—a direct verification that demodulation is the sigmoid-curvature term of Eq. (10). The closed-loop JR operating points (dots) sit on the lobe where σ'' is large and negative—not by design, but because that is where the JR resting point lies as the alpha Hopf is approached (the saturating upper flank of the sigmoid). The mechanism is sign-agnostic: σ'' sets only the sign/phase of the demodulated response while $|\sigma''|$ sets its gain, so the entire resonance picture is unchanged (up to an overall sign) in the $\sigma'' > 0$ lobe, which this open-loop sweep already spans. The sign flip itself is a falsifiable prediction (§4.10).

log slope 1.99 against the predicted 2), and replacing the field-receiving sigmoid by its linearization collapses the response by 712× (1.40 → 0.0020 mV)—the nonlinearity is the detector, and its curvature is the gain.

4.5 Two native resonances and nonlinear Arnold tongues in the LaNMM

The single JR column isolates and lets us analyze the mechanism; the laminar model (LaNMM, §2.1) confirms it in a cortical column with its *own* two resonances. Driving one population with the AM input of Eq. (15) and reading the alpha-band power of the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials (Methods, Fig. S2), we recover the same envelope resonance, now carrying the column's intrinsic rhythms.

Driving the deep P_1 (alpha) loop directly and sweeping the envelope frequency Δf reproduces the single-column picture in the laminar model (Figs. S9, S3). With the P_1 input as the bifurcation parameter, the stable-focus side (high drive) gives a forced alpha resonance at ∼10 Hz that grows as the alpha Hopf is approached, while the limit-cycle side (low drive) entrains the autonomous alpha—the same stable-focus/limit-cycle pair as the JR column (Fig. 4), now read out as the deep-pyramidal alpha lock-in. The heuristic laminar model thus shows the JR resonance *and*, in addition, the carrier-dependent Arnold tongues that follow.

Sweeping the modulation amplitude A and beat frequency Δf at a fixed carrier produces *nonlinear Arnold tongues*: alpha power in P_1 is maximal when Δf matches the deep alpha (∼10 Hz) and grows with A (Figs. 7a,b and S10a,b)—envelope resonance and phase locking of the slow generator to the recovered envelope, exactly as in the JR resonance curves but now in a biophysical two-band column.

The carrier dependence reproduces, and physically grounds, the two coupling limits of §2.2–§2.5. Sweeping carrier f_c against beat Δf: when the AM input is delivered to the *superficial* loop (P_2), strong P_1 demodulation requires a *double* resonance—f_c near the P_2 gamma (∼40 Hz) *and* Δf near the P_1 alpha (∼10 Hz) (Fig. 7c,d): the fast loop acts as a tuned "RF front end" that must

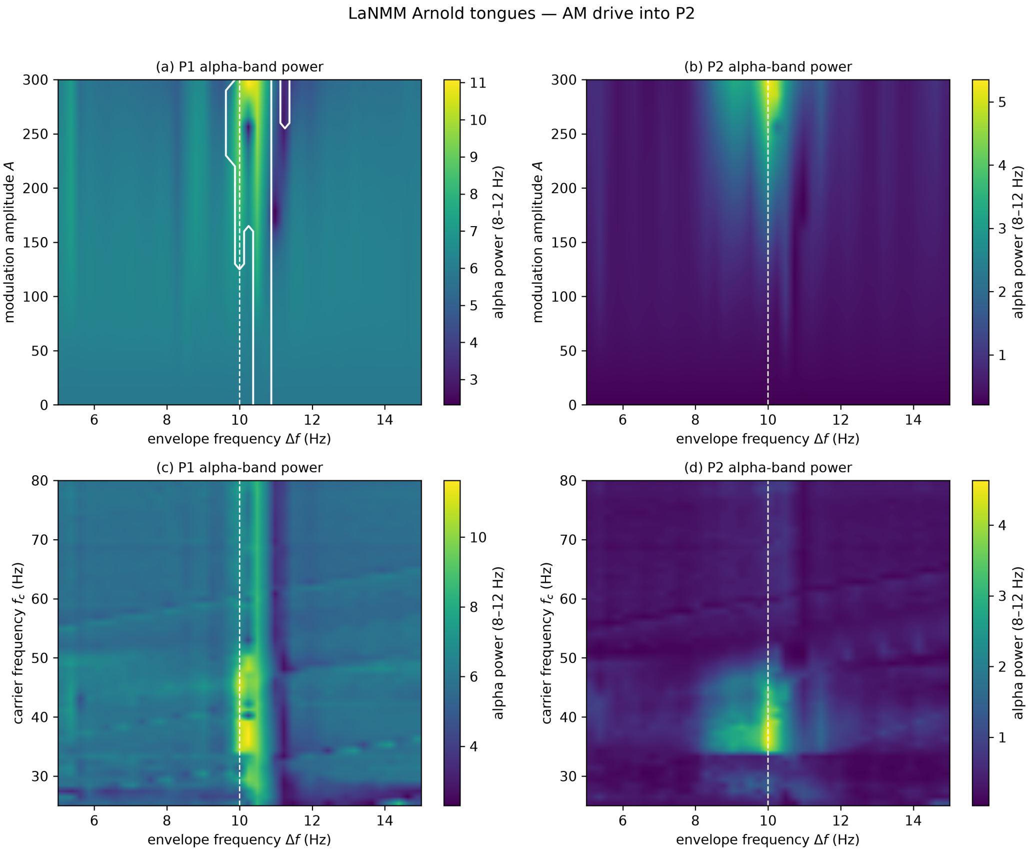

Figure 7: LaNMM Arnold tongues: AM input to the superficial P_2 (gamma) loop. *(a,b)* Alpha-band power of P_1 and P_2 vs. modulation amplitude A and beat frequency Δf (carrier 40 Hz), with elevated power around the P_1 alpha (∼10 Hz, dashed line). White contour in *(a)*: the 1:1 frequency-captured set, $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix C), which is what defines the tongue. Band power alone cannot separate capture from a forced response, since both place power at Δf; the captured set is therefore computed from the dominant output frequency and drawn explicitly. Its width grows monotonically with drive, from ≈ 0.6 Hz at $A \to 0$ (the tolerance-limited tip, $2 \times 0.3 = 0.6$ Hz) to 1.75 Hz at $A = 300$. *(c,d)* Alpha power vs. carrier frequency and Δf at $A = 250$: P_1 demodulation requires *both* a carrier ≈40 Hz (P_2 gamma) and Δf ≈10 Hz (P_1 alpha)—a double resonance.

itself be excited before it can pass a demodulated envelope to P_1. When the input is delivered *directly* to the slow generator (P_1), the response is broadband in f_c—a vertical ridge at $\Delta f \approx 10$ Hz across the whole carrier range (Fig. S10c), i.e. *carrier-independent*, just as Eq. (10) predicts for direct coupling. Real TI, in which the field reaches deep generators through faster elements, sits between these limits: a carrier matched to a fast (gamma) network resonance couples best, but the envelope response persists—weaker—off resonance. This both confirms the JR mechanism and refines it: the "inert carrier" of §2.3 is the direct-coupling idealization, while a band-limited intermediary makes the carrier choice matter, consistent with frequency-tagging/intermodulation paradigms [30].

Concretely the LaNMM predicts that TI efficacy should (i) peak when Δf matches a region's intrinsic alpha; (ii) widen and strengthen with modulation amplitude (the tongue); and (iii) for fields delivered via superficial fast circuits, be largest when the carrier is matched to a gamma network resonance—e.g. two nearby gamma carriers summing to a ∼40 Hz-centered AM field—while remaining present, weaker, for off-resonance carriers [31, 32].

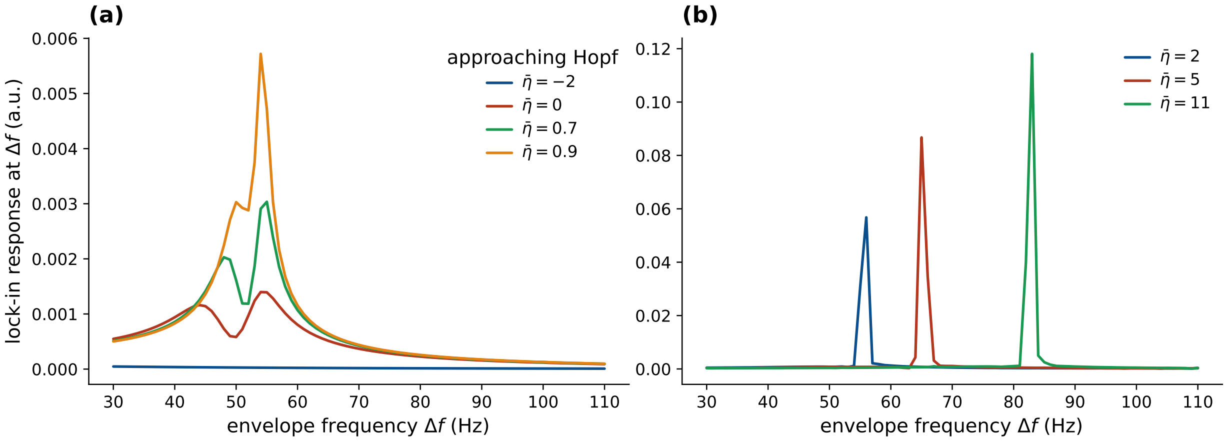

Figure 8: NMM2 PING gamma resonance (exact mean field). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under an AM field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_{\mathrm{Hopf}} \approx 1$) forced resonance at the gamma frequency ($f_0 \approx 55$ Hz), growing as the Hopf is approached ($1/\gamma$ gain), with the resonant frequency rising with $\bar{\eta}$. *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_{\mathrm{Hopf}}$) entrainment of the autonomous gamma. The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity.

4.6 Exact mean-field confirmation: the NMM2 PING gamma generator

The JR column and the LaNMM both use the heuristic firing-rate sigmoid (NMM1). As a third, independent test we move to the exact next-generation mean field (NMM2, Eq. (3)), where the demodulating nonlinearity is the *derived* quadratic v^2 term rather than a postulated curve. We drive the two-population E–I PING motif of Eq. (16) (Methods)—a gamma generator that loses stability through a supercritical Hopf (at $\bar{\eta} \approx 1$, $f_0 \sim 55$ Hz), the gamma counterpart of the JR alpha Hopf—with the AM field entering the excitatory current, so the carrier is rectified by the exact v_E^2 term.

Driving with an AM field (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) and locking in the excitatory rate r_E at Δf reproduces the whole picture (Figs. 8, S11). Below the Hopf the response is a forced resonance at the gamma frequency whose height grows as the bifurcation is approached—the $1/\gamma$ gain of Eq. (11); above it the demodulated drive entrains the autonomous gamma. The two-dimensional map traces a ridge at $\Delta f \approx f_0$ that intensifies through the Hopf. Unlike the static-sigmoid models, the resonant frequency itself *shifts with the operating point*—the ridge is diagonal, not vertical—a direct signature of the dynamic, state-dependent transfer of the exact theory [19]. That the same demodulation-and-near-Hopf-resonance mechanism appears with the nonlinearity *derived* from QIF spiking dynamics, rather than assumed, shows it is not an artifact of the heuristic sigmoid.

4.7 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input p, but p also shifts the operating point, entangling detection (σ'') with amplification. The sharper, mechanistic statement is that the *network coupling* sets the gain on its own. In the JR column we vary the global connectivity C—the coupling strength J—and, for each C, solve analytically for the drive $p(C)$ that *pins* the operating point v^*, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\mathrm{open}} = \frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ are held exactly constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows $\sim$27$\times$ and the envelope resonance sharpens (Fig. 9a,b)—entirely a network effect, since detection is frozen and only γ changes. The recovered gain $A_\Omega / A_{\mathrm{open}}$ tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input p rather than the coupling C yields the same gain–γ relation, so the amplification depends only on the distance to criticality, not on which parameter delivers it. All

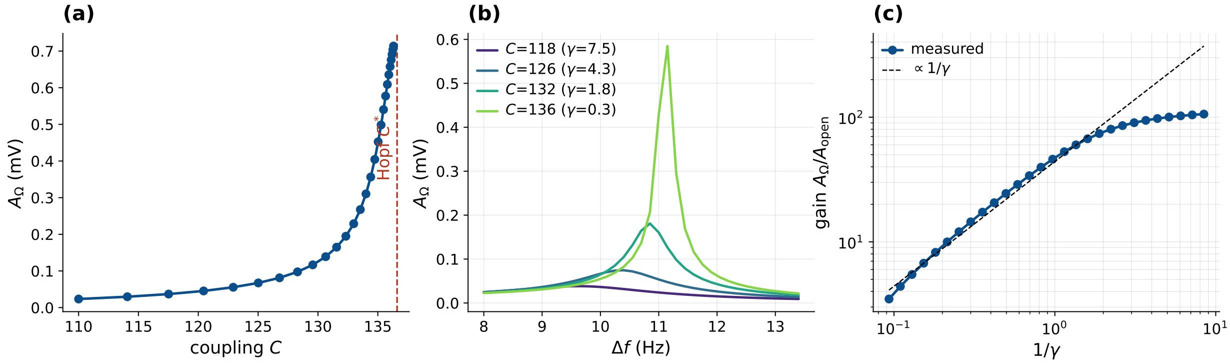

Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit). The operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling $C\,(= J)$ approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain A_Ω/A_{open} follows $1/\gamma$ (dashed) and saturates near the bifurcation.

these sweeps stay on the stable-focus side ($\gamma > 0$), approaching the supercritical Hopf from below: the $1/\gamma$ growth is the forced-resonance susceptibility of a damped network, not entrainment of an autonomous cycle. On the other side of the bifurcation the column oscillates on its own, and the demodulated drive instead phase-locks that limit cycle within an Arnold tongue (Fig. S5; also the entrainment side of Fig. 4 and the LaNMM and NMM2 maps)—1:1 frequency capture over a band that widens with drive, quantified by the locking range rather than a diverging $1/\gamma$ gain, the two coinciding only at onset. Because the drive is weak, on the autonomous side it biases phase (timing) more than amplitude, consistent with the timing-not-rate signature (§4.8).

This becomes exact in the next-generation theory. In the NMM2 PING the coupling C is the *literal* mean inter-neuron (QIF) synaptic weight J, and the demodulating nonlinearity is the exact v_E^2 term, whose quadratic coefficient is a fixed constant—so detection needs no pinning. Sweeping J at fixed background drive $\bar{\eta}$ toward the gamma Hopf J^*, the demodulated peak of r_E grows as $1/\gamma$ and saturates at criticality (Fig. S13): the heuristic JR result, reproduced with the coupling *and* the rectifier both derived rather than assumed, and a direct counterpart of the coupling-controlled weak-field sensitivity of QIF networks [19]. The two columns realize one band-agnostic law in their two native bands—alpha (JR) and gamma (NMM2): the single-cell nonlinearity detects the envelope (coarse-grained into σ'', or the exact v^2), while the sharp, tunable gain is the network's—a resonant amplification set by proximity to the Hopf and adjustable through the coupling.

4.8 The resonance amplifies envelope timing, not the mean rate

The demodulator feeds the network a baseband term and a line at the beat Ω (Eq. (9)), and the transfer $G(\Omega)$ of Eq. (11) treats them differently: $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $G(0) \propto 1/\omega_0^2$ stays flat. Pinning v^* to hold detection fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, the AC lock-in at the resonant beat $\Delta f = f_0$ grows $\sim$17×, tracking $1/\gamma$ before saturating near criticality, while the DC shift of the mean pyramidal firing rate stays within a few mHz of zero (Fig. S14). The network amplifies the envelope-locked AC response and not the DC rate, so a flat mean rate is the expected signature even where TI is acting—the mesoscopic counterpart of the observation that TI alters spike timing but not firing rate in the primate brain [6].

4.9 Microscale confirmation: TI realigns spike timing in the QIF network

The mesoscale prediction has a direct microscale test, since NMM2 is the *exact* mean field of an all-to-all network of quadratic integrate-and-fire (QIF) neurons with Lorentzian-distributed excitabilities [17, 11, 19]. Integrating that spiking network ($N = 8000$ neurons per population, parameters identical to the PING mean field; Methods) shows TI bunching spike *times* into the recovered envelope without changing the mean rate, on both sides of the gamma Hopf (Fig. 10). Below the Hopf (forced) the field-free network fires asynchronously at a flat rate; with TI on, spikes condense into envelope-locked bands (Fig. 10a,b). Above the Hopf (oscillatory) the free-running gamma is *gated* by the TI beat—forced cross-frequency modulation (PAC), not frequency entrainment, since the ≈11 Hz detuning lies outside the 1:1 tongue: its bursts group into the $\Delta f = 42$ Hz envelope while the intrinsic ∼53 Hz gamma persists (Fig. 10c,d). Because the two rhythms coexist in a nonlinear medium, the sigmoid mixes them into intermodulation sidebands in the population rate—most prominently the direct Δf line and the sum component near 95 Hz (53+42), with the difference tone near 11 Hz (the detuning itself); this is the spiking-network fingerprint of the same nonlinear frequency mixing that operates at the macroscale, and it is the sideband signature of *gating* rather than the frequency pulling of entrainment. The spiking network reproduces the exact NMM2 mean field (matched gamma spectrum and mean rate, Fig. S15a), and across both regimes the mean rate is unchanged (×1.05–1.10) while the Δf modulation depth of the population rate rises from near zero to 0.19–0.76 (Fig. S15c): TI realigns timing, not rate—the spiking-level counterpart of §4.8 and a direct microscale analog of the in-vivo spike-timing-without-rate finding [6].

4.10 Falsifiable predictions and tests

The mechanism makes three sharp predictions, each with a clean experimental handle.

1. **Carrier independence at fixed polarization.** Once the post-coupling polarization $\varepsilon = \kappa(f_c)E_0$ is matched, the envelope response no longer depends on f_c, even though the *applied*-field threshold rises with f_c through the membrane roll-off (Eq. (13))—as single-neuron accounts also predict [4, 3]. *Test:* hold the in-zone polarization fixed (raising E_0 to compensate the measured $\kappa(f_c)$) while sweeping f_c; the network signature is a flat envelope response.
2. **Operating-point (PEIX) control and sign flip.** Because $A_\Omega \propto \sigma''(v^*)$, the response scales with the curvature at the operating point and *reverses sign* across the inflection, vanishing at v_0 where the gain σ' (PEIX, Eq. (12)) peaks. *Test:* shift v^* with tonic drive, neuromodulation, or task state and read out the predicted scaling and sign reversal of the TI response.
3. **Frequency selectivity and state-dependence.** The response is largest when Δf matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation, so manipulations that move the operating point or the distance to the Hopf—psychedelics, PV-interneuron dysfunction, attention, arousal—should scale, and can abolish or sign-reverse, TI efficacy [9]. *Test:* sweep Δf through a region's intrinsic band across brain states and track the resonant peak and its width.

Table 2 confronts these predictions—together with two corollaries the mechanism already shares with published observations—against existing evidence. Two are, in qualitative form, *already supported*: TI alters spike timing but not firing rate in vivo [6] (our $G(\omega_0) \gg G(0)$ asymmetry, §4.8–4.9), and most pyramidal TI responses vanish when synaptic transmission is blocked [7] (a network, not single-cell, read-out). A quantitative reanalysis of these datasets against the model is a natural next step.

5 Discussion

We have given a mesoscopic, population-level account of TI demodulation in which the firing-rate sigmoid is a square-law detector and proximity to a Hopf bifurcation turns the recovered envelope into a sharp, frequency-selective response. This complements rather than competes with the single-neuron ion-channel rectification account [3]: both are rectify-then-filter demodulation, but ours needs no specific channel, lives at the scale of the EEG-generating population, and inherits the brain's own resonances, so a region's tuning is set by its intrinsic dynamics just as a radio is tuned

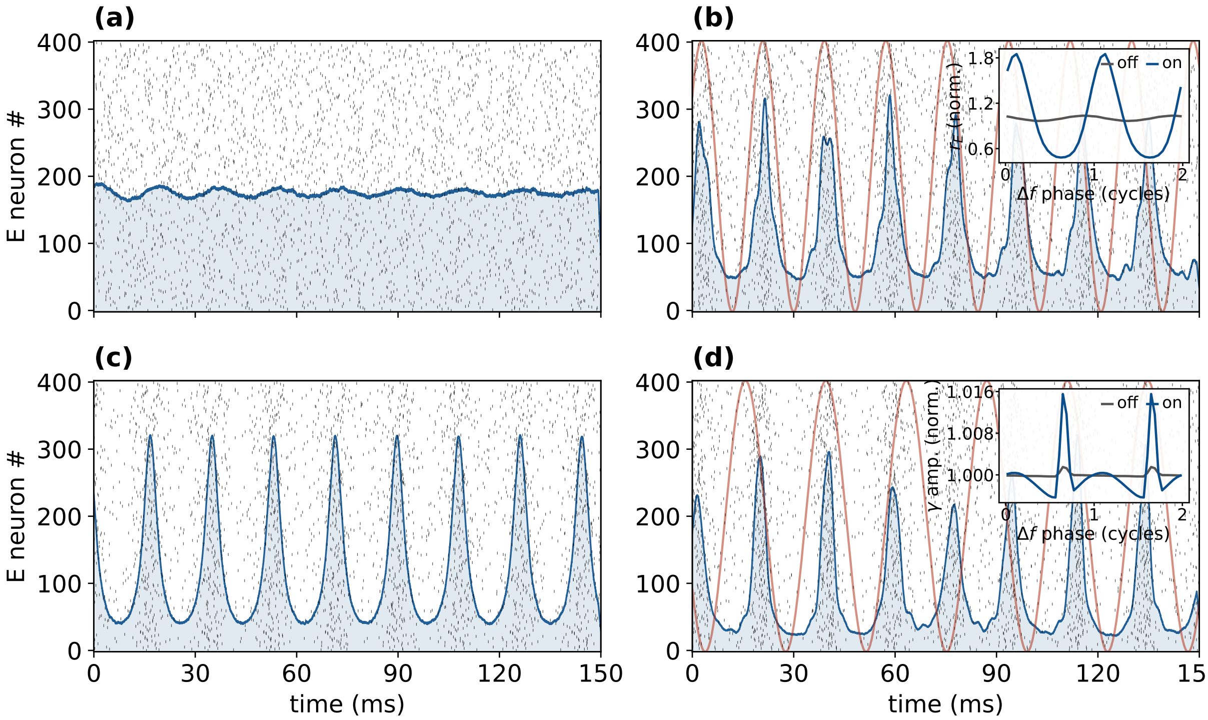

Figure 10: QIF spiking network (microscale of the NMM2 PING *gamma* generator): TI realigns spike timing, not mean rate. $N = 8000$ excitatory and inhibitory quadratic integrate-and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (16)). This circuit's intrinsic rhythm is *gamma* (∼53 Hz), so, as in the alpha figures but one band up, the TI beat Δf is set in the gamma range (here much larger than the $\Delta f \sim 11$ Hz of the Jansen–Rit alpha column; carrier $f_c = 300$ Hz). Black ticks are E spikes (400 of 8000, a representative sample), blue the population rate $r_E(t)$, red the TI envelope; the boxed value is the Δf modulation depth $A_\Omega/\langle r_E\rangle$ of the rate (near zero off, large on). The bifurcation parameter is the common background drive $\bar{\eta}$ (gamma Hopf at $\bar{\eta}_{\text{Hopf}} \approx 1.1$). *(a)* Forced ($\bar{\eta} = 0.85$, just below the Hopf—a damped gamma resonator), TI off: asynchronous firing, flat rate. *(b)* Forced, TI on, driven *near* the gamma resonance ($\Delta f = 55$ Hz): spikes bunch into the envelope (modulation depth 0.76). *(c)* Oscillatory ($\bar{\eta} = 1.5$, above the Hopf—an autonomous ∼53 Hz gamma cycle), TI off: free-running gamma, no Δf lock-in. *(d)* Oscillatory, TI on, with the beat set *below* the autonomous gamma (beat $\Delta f = 42$ Hz, a detuning of ≈11 Hz from $f_0 \approx 53$ Hz): the TI beat *gates* the gamma—the population rate gains a Δf-locked component (modulation depth 0.19) while the intrinsic ∼53 Hz gamma persists. This is forced cross-frequency *gating* (PAC), *not* frequency entrainment: the ∼11 Hz detuning places the beat outside the 1:1 Arnold tongue (contrast the genuine frequency locking of Fig. S5), so the gamma is amplitude-modulated at Δf rather than pulled to it—demodulation surviving above the Hopf as envelope-to-rhythm coupling. The detuning is what certifies the Δf-locked component as a real driven effect rather than coincidence. *Insets* (each normalized to its own mean, so the y-axis reads the relative modulation; x-axis is the Δf beat phase in cycles; TI off gray, on blue): *(b)* the population rate folded on the beat phase (the forced response *is* at Δf); *(d)* the gamma-band *amplitude* folded on the beat phase—taking the amplitude removes the gamma carrier, so the near-commensurate 53/42 Hz gamma cannot contaminate the fold (a raw-rate fold would). Both are flat with TI off and show a clear Δf-locked modulation with TI on, making the envelope-locked timing direct where the raster alone leaves it subtle.

by its tank. Because each mass is $\sim 10^4$ neurons, demodulation is a *population* read-out by the sigmoid—a single-neuron nonlinearity coarse-grained over the ensemble—while the amplification (the near-Hopf resonance) is the emergent *network* property; the single cell, beyond contributing to the sigmoid's shape, need only supply a fast nonlinearity that samples the carrier.

This factorization identifies where the carrier-sampling nonlinearity may live. The TI literature points to three best-supported, non-exclusive loci. (i) *Axonal Na^+ channels* at nodes and the AIS: fast voltage-gated Na^+ rectifies because activation outpaces inactivation [3, 33], Na^+-inactivation conduction block passes the slow envelope while removing the carrier [3, 4], and persistent Na^+—a

Prediction (model basis)	Existing evidence	Sharp test
Timing, not rate: TI amplifies the envelope-locked AC, not the DC mean rate ($G(\omega_0) \gg G(0)$)	*Supported in vivo*: TI alters spike timing but not firing rate, primate [6]	Lock-in of spike timing at Δf vs. the mean-rate change
Network, not single-cell, read-out (population sigmoid + synaptic recurrence)	*Supported in vitro*: most pyramidal TI responses vanish under synaptic block [7]	Synaptic block should abolish the demodulated response
Carrier independence at fixed polarization (no ω_c in A_Ω)	TI thresholds inherit the carrier's f_c-dependence [4, 5]; bare kHz weakly effective [29]	Fix in-zone polarization, sweep $f_c \to$ flat envelope response
Operating-point (σ'') control and sign reversal across v_0	Cell-type-specific TI responses (PV vs. pyramidal) [7]	Shift v^* (neuromodulation/state) $\to$ scaling and sign flip
Frequency selectivity, sharpened $\sim 1/\gamma$ near criticality	— (a sharp new prediction; largely untested)	Sweep Δf through a region's band across brain states

Table 2: The mechanism's predictions confronted with existing in-vivo / in-vitro evidence. Entries cite documented qualitative findings; quantitative reanalysis is left to future work.

fast subthreshold rectifier tied to axonal resonance—could even set an optimal Δf. (ii) *Presynaptic terminals*, which polarize $\sim$2–10× more than somata [34, 35, 36] and convert depolarization into transmitter release through the steeply nonlinear Ca^{2+}–release relation. Sensitivity is not speed, however. For an isopotential compartment $\tau_m = R_m C_m$ with both quantities taken per unit area, so the membrane time constant does not shrink with compartment size: a small bouton has a small capacitance and a correspondingly large input resistance. A terminal therefore tracks a kHz carrier no better than a soma does, and the short time constants available in axons come from high-conductance nodal and myelinated membrane rather than from small size. Terminals are thus the *sensitive* locus, not the fast one, and the carrier sampling the factorization requires is supplied by (i). (iii) *Threshold spiking*, where near threshold the spike probability is a smooth nonlinearity and a weak bias is rectified through stochastic resonance [24, 23]. In model cortical neurons TI produces block, onset and phasic responses depending on geometry [33], and in vivo it alters spike timing rather than rate in the primate brain [6]—consistent with the timing-not-rate signature of our resonance (§4.8). The mesoscopic mechanism is agnostic to which of these does the sampling.

The same fast-element logic explains TMS, and underscores that high-frequency drive is not somatic. TMS pulses carry kHz spectral content (peak $\sim$3.9 kHz), which a $\tau_m \approx 16$ ms soma attenuates by $\sim$50 dB; TMS—like TI—therefore cannot be driving the soma at kHz, and works instead through fast excitable elements whose chronaxie ($\sim$0.15–0.25 ms) sets an effective corner near $\sim$800 Hz [37, 38]. That the bare unmodulated kHz carrier is itself weakly effective on network oscillations [29] is the complement of the same fact: the carrier matters only where a fast nonlinearity rectifies it.

The amplification half is not specific to TI. A conventional tACS field oscillates in the envelope band directly, so it needs no demodulation—but it drives the same near-Hopf resonator and should inherit the same network gain: a tACS rhythm at Δf near a region's natural frequency is resonantly boosted as the network approaches criticality ($\propto 1/\gamma$) and as its coupling J grows, the Arnold-tongue picture of weak-field stimulation [19]. We confirm this in the JR column (Fig. S4): a direct Δf drive, swept in coupling toward the alpha Hopf at fixed operating point, reproduces the same $1/\gamma$ growth and resonance sharpening as the demodulated TI drive, but linear in the field rather than square-law. TI thus differs from tACS only by a sigmoid demodulation front-end that turns a kHz carrier's envelope into an effective in-band drive; downstream, both share the criticality- and coupling-controlled amplifier, so—matched for effective in-band drive—TI and tACS should entrain alike and show the same state- and coupling-dependence.

The mechanism's falsifiable predictions (carrier independence at fixed polarization, the PEIX-controlled sign flip, and frequency-selective state-dependence) are consolidated with their tests in §4.10; they could be read out in the same models already applied to Alzheimer's disease and psychedelics [9]. The two-band LaNMM results (§4.5) bear this out: its JR (alpha) and PING (gamma) subcircuits give the same sigmoid demodulation two native resonances, with the beat frequency selecting the band; this embeds TI demodulation in the HAM read-out hierarchy [10], where envelope extraction is the inverse of the amplitude-modulation encoding used for prediction-error evaluation. Concretely, the most effective beat tracks the target's own rhythm—$\Delta f \sim 10$ Hz to engage an alpha generator, $\Delta f \sim 40$–55 Hz for a gamma (PING) circuit, as in the NMM2/QIF figures—so TI's optimal Δf is a testable function of regional spectral anatomy rather than a single universal value, and gamma-band targets call for correspondingly higher beat frequencies.

Amplification is critical slowing, of a mean-field kind. The network effect at the heart of this work, gain $\propto 1/\gamma$ as the damping γ (the leading Jacobian eigenvalue) vanishes at the Hopf, is the dynamical signature of *critical slowing*: a susceptibility that diverges as a system approaches a bifurcation. Whether that warrants the word "criticality" depends on the model. In the heuristic Jansen–Rit and LaNMM columns the Hopf is a bifurcation of a finite-dimensional, fitted ODE, a near-bifurcation statement rather than a phase transition. In the exact next-generation mean field (NMM2/MPR) it is sharper: the reduction from the spiking QIF population is exact in the thermodynamic limit, and the dynamical variables (r, v) are genuine *order parameters*, conformally related to the Kuramoto–Daido order parameter through the Ott–Antonsen ansatz [39, 17], so the Hopf is a true macroscopic transition of the underlying many-body system and the $1/\gamma$ gain is its diverging susceptibility. One qualifier keeps this honest. Because the reduction assumes all-to-all coupling, the critical point is *mean-field*: it delivers temporal critical slowing but no diverging *spatial* correlation length, so the column realizes the temporal-susceptibility face of criticality and not the scale-free spatial one. With the inter-neuron coupling J as the control parameter (Fig. S13), TI drives a population toward a genuine, if mean-field, critical point and reads out its susceptibility, and its efficacy is largest where the network sits closest to that transition.

Prior work already places the operative nonlinearity in the network. Large-scale simulations attribute tACS entrainment to the recurrent excitatory coupling of pyramidal cells and recover an Arnold tongue from it [40]; the dynamical transfer function of an E–I network, computed for both rate and spiking models, shows a resonance whose position depends on recurrent coupling [41]; and a cortical model driven by an amplitude-modulated high-frequency waveform phase-locks to the envelope through the same target engagement as low-frequency tACS [42]. The broader case for treating stimulation as interacting with ongoing dynamics rather than imposing on passive tissue has been made explicitly [43]. What we add is the specific pair of factors: σ'' as the square-law detector, with the demodulated amplitude in closed form, and $1/\gamma$ as the amplification, with its size tied to a measurable relaxation time.

The amplitude budget does not close at human field strengths. A transcranial field must clear two hurdles before it can produce a response at Δf, and they need different answers.

The first hurdle is that weak fields barely move a membrane. Measured coupling is 0.12–0.21 mV of somatic polarization per V/m in hippocampus [44, 45], and up to ∼0.5 mV per V/m for cortical L5/6 pyramidal cells [46]. Human cortex sees about 0.8 V/m at 2 mA [47, 48], and most of the applied current is shunted through scalp and skull before it arrives [49]. So the field shifts a soma by a few tenths of a millivolt, against the ∼23 mV that separates rest from threshold [50]. No cell fires because of the field alone. This is as true of tACS as of TI.

The second hurdle belongs to TI alone. Its field carries no power at Δf, so the response there has to be manufactured, and that is what σ'' does.

The two answers do not substitute for each other. The resonance clears the first hurdle: a population near its own bifurcation responds to a weak push far more strongly than an isolated cell. It cannot clear the second, because it amplifies TI and tACS alike and cancels when the two are compared.

The resonance buys a factor of order ten. Writing the reverberation time of the weakly damped focus as $\tau = 1/\gamma$, the gain at resonance relative to the same circuit's response to a steady input is $\mathcal{Q} = \omega_0/2\gamma = \pi f_0 \tau$: π times the number of cycles the ringing survives. Cortical population activity in vivo relaxes over a few hundred milliseconds [51], which at $f_0 = 10$ Hz gives $\mathcal{Q}$ of order 10. That measured timescale is wideband rather than alpha-specific, so it sets the order of magnitude and not the value. Gain also costs bandwidth: the resonance is $1/(\pi\tau)$ wide, about 1 Hz, so a Δf fixed at a group-mean alpha will drift outside it during a session.

At equal field strength, TI is far weaker than tACS. A tACS field oscillating at Δf drives the population directly. TI must first manufacture its Δf signal out of a kHz carrier, and that conversion is inefficient. With per-carrier polarizations $\varepsilon_{1,2}$ the demodulated drive is $\frac{1}{2}\sigma''\varepsilon_1\varepsilon_2$, against $2\sigma'\varepsilon$ for tACS at the same peak polarization. Their ratio, $\eta = \sigma''\varepsilon/4\sigma'$, is proportional to ε: a square-law detector gets *less* efficient as the signal gets weaker. Formally η reaches unity at $\varepsilon \approx 7$ mV, but that is the same order as the distance to threshold and lies outside the range where the expansion itself holds, so parity with tACS is not reachable by this route rather than merely expensive. At 1 V/m with a 1 kHz carrier, TI comes out $\sim$300$\times$ weaker than tACS if the carrier reaches a fast nodal compartment, and $\sim 10^6\times$ weaker if it reaches only the soma. The resonance does not make this back, because it multiplies both equally.

What TI buys for that price is targeting: a field applied directly at Δf cannot be steered to a deep structure, whereas two kHz beams can.

The same nonlinearity explains why TI works in rodents and disappoints in humans. A mouse brain sees $\sim$383 V/m from a TI montage, while optimized human montages reach 0.24–0.57 V/m at depth [52], a field ratio of order 10^3. A nonlinear detector amplifies that disparity rather than preserving it, but not by the full square. At mouse field strengths the per-carrier polarization arriving at the detector is $\varepsilon \sim 18$ mV, roughly 80% of the way from rest to threshold, so the sigmoid saturates and the small-signal expansion (10) no longer holds. Evaluating the full sigmoid rather than its quadratic term gives a drive ratio of order 10^4, where extrapolating the square law would predict 10^6. The rodent experiment is therefore not the human experiment scaled down: the mouse is driven into saturation, while the human, at $\varepsilon \sim 0.02$ mV, sits deep in the perturbative regime where the quadratic penalty is worst. A network model finding that TI needs $\sim$100$\times$ the tACS amplitude [53] is consistent with a nonlinear detector operating there.

What survives is the mechanism, not a solution to the dose problem. It explains TI's phenomenology and the rodent-to-human collapse, and names the three quantities worth improving: the carrier's access to a fast compartment, the operating point through σ'', and proximity to the bifurcation through $\mathcal{Q}$.

Several limitations should be kept in view. The field enters the population sigmoid directly; we treat the field-to-nonlinearity coupling $\kappa(f_c)$ phenomenologically (Eq. (8)) rather than modeling channel kinetics, and ε should be read as the post-coupling polarization, not the applied field (§2.2). The population sigmoid is likewise a coarse-grained stand-in for distributed single-cell nonlinearities. JR alpha near this Hopf is high quality-factor, so the stable-side peaks are narrow and would be broadened by noise and population heterogeneity. The 100 Hz carrier is below TI's kHz range, but this is a testable regime in its own right (§2.5) and the argument depends only on a timescale separation. Finally, the LaNMM realization here still treats a single, deterministic, noiseless column with a phenomenological field coupling; coupled noisy columns and an explicit, realistic $\kappa(f_c)$ remain natural next steps. The human evidence is also not uniformly positive: a controlled study found peripheral but no central effects [54], and a systematic review reports that occipito-parietal TI did not modulate alpha power [55]. The narrowband prediction above turns the second into a design question, namely whether Δf tracked individual alpha closely enough to stay inside a $\sim$1 Hz response band; that is a prediction to be tested, not an explanation to be assumed.

6 Conclusion

A population sigmoid and a second-order-synapse network near a Hopf bifurcation suffice to demodulate a temporal-interference field and resonate at the recovered envelope frequency. The sigmoid's curvature is inherited from single-neuron nonlinearity, coarse-grained over the $\sim 10^4$ neurons of the mass; what is emergent is that demodulation is read out at the *population* level, needing no specific channel, and that the resonant amplification is a *network* property. A fast single-cell nonlinearity (Na^+ at nodes/AIS, persistent Na^+, presynaptic terminals) need only sample the carrier. The column then behaves as a tuned AM receiver—a nonlinear detector followed by a resonant tank—so that *wherever a fast-enough nonlinearity sees the carrier, the population sigmoid demodulates and the network amplifies.*

The amplification half is not automatic: the gain scales as $1/\gamma$ with the distance to the Hopf, so a large, sharply tuned response appears only near criticality. It is tunable both by brain state and by the coupling itself—holding detection fixed, the demodulated gain grows toward $1/\gamma$ as the network is driven to criticality by its own connectivity (§4.7), in the heuristic JR column and, with coupling and rectifier both derived, in the exact NMM2 mean field. Both halves are thus **state-dependent**—detection through $\sigma''(v^*)$ (which changes sign across the inflection), amplification through γ—set by neuromodulation, attention, arousal and pathology (PEIX). TI efficacy is then as much a property of *brain state* as of the stimulus: largest near criticality, and tunable—even abolished or sign-reversed—by shifting the operating point.

This picture is both testable and actionable. It makes sharp, falsifiable predictions (§4.10)—carrier-independence at fixed polarization, a sign reversal of the response across the sigmoid inflection, and a response that peaks when Δf matches a region's intrinsic rhythm—and it argues that TI should be dosed to brain state rather than delivered open-loop. More broadly, it places external stimulation inside the same framework as the cortex's handling of its *own* rhythms: if a column is an AM *transceiver* for endogenous cross-frequency coupling [9, 10], TI simply drives it as a *receiver* for an external carrier. The natural next steps—a realistic field-to-nonlinearity coupling $\kappa(f_c)$ and coupled, noisy columns—point toward a quantitative, state-aware account of who responds to TI, and when.

Code and data availability

All simulation and figure-generation code, the committed datasets, and a one-command reproduce script (`run_all.py`) are openly available at `https://github.com/giulioruffini/TI-demodulation-paper`, archived at Zenodo (doi:10.5281/zenodo.21009618). The repository `README` documents the per-figure script mapping (a figure→script table) and the dependency order, so every figure in this note can be regenerated from the committed inputs. The code is released under CC-BY-4.0.

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Competing interests

GR is Chief Scientific and Technology Officer of Neuroelectrics, which develops and commercializes transcranial stimulation devices. BM, AJ and FC are affiliated with Neuroelectrics. The remaining authors declare no competing interests.

Author contributions

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

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A Notation and units

These neural-mass models are phenomenological: potentials are in mV, firing rates in s^{-1} ($\equiv$Hz), time constants in ms, and frequencies in Hz, unless noted; a "–" marks a dimensionless quantity. A few glyphs are reused across the four models (JR, LaNMM, NMM2/MPR, QIF) by long-standing convention; these are disambiguated by context and flagged below. In particular, A is the excitatory PSP *gain* in the JR/LaNMM synapses, while in the LaNMM *drive* $e(t)$ and (as A_f) in the NMM2/QIF field it is the applied-field modulation *amplitude*—the counterpart of the single-column ε. Likewise C is the JR connectivity constant and the NMM2 global synaptic gain C ($\equiv J$).

Symbol	Meaning	Units
Firing-rate nonlinearity (sigmoid)		
$\sigma(v)$	population firing-rate transfer function	s^{-1}
$\sigma'(v)$, $\sigma''(v)$	sigmoid slope and curvature; σ'' sets the demodulation gain	$s^{-1}mV^{-1,-2}$
e_0	half the maximal firing rate ($\sigma \to 2e_0$ as $v \to \infty$)	s^{-1}
v_0	sigmoid inflection (half-maximal) potential	mV
ρ	sigmoid slope parameter	mV^{-1}
v^*	operating point (field-free fixed point of v)	mV
$v = y_1 - y_2$	net pyramidal membrane potential (LFP proxy)	mV
PEIX	Population Excitation–Inhibition Index (a σ'-based brain-state index)	–
Single-column Jansen–Rit		
A, B	excitatory, inhibitory PSP gains	mV
a, b	excitatory, inhibitory synaptic rates (inverse time constants)	s^{-1}
y_1, y_2	excitatory, inhibitory post-synaptic potentials	mV
C, $C_1 - C_4$	intracolumn connectivity constants	–
p	external input rate; JR bifurcation parameter	s^{-1}
Applied TI field and coupling		
$s(t)$, $e(t)$	applied AM field perturbation (JR; LaNMM)	mV
ε / A / A_f	field modulation amplitude (JR / LaNMM drive / NMM2–QIF)	mV
m	modulation depth	–
f_c, $\omega_c = 2\pi f_c$	carrier frequency	Hz, $\mathrm{rad\,s^{-1}}$
f_1, f_2	the two applied carrier tones ($f_c = (f_1 + f_2)/2$)	Hz
Δf, $\Omega = 2\pi\Delta f$	envelope (beat) frequency, $\Delta f = f_2 - f_1$	Hz, $\mathrm{rad\,s^{-1}}$
$E(t)$, $\delta V(t) = \lambda E$	applied field; induced membrane perturbation	$\mathrm{V\,m^{-1}}$; mV
λ	lumped (polarization-length) field-to-membrane coupling	$\mathrm{mV / (V\,m^{-1})}$
$\kappa(f_c)$	applied-field-to-polarization coupling	$\mathrm{mV / (V\,m^{-1})}$
$H_m(\omega)$, τ_m	somatic membrane low-pass and its time constant	–; ms
τ	fast-element (node/AIS/terminal) time constant	ms
Network dynamics, response, and bifurcation		
$\lambda_\pm = -\gamma \pm i\omega_0$	leading Jacobian eigenvalue pair	s^{-1}
γ	damping of the critical mode ($\propto$ distance to the Hopf)	s^{-1}
ω_0, $f_0 = \omega_0/2\pi$	natural (Hopf) frequency	$\mathrm{rad\,s^{-1}}$, Hz
$\mathcal{Q} = \omega_0/2\gamma = \pi f_0 \tau$	resonance quality factor ($\tau = 1/\gamma$ the reverberation time)	–
$G(\Omega)$	network resonance gain (Lorentzian, $\propto 1/\gamma$ at ω_0)	–
A_Ω	demodulated response: lock-in amplitude of v at Ω	mV
A_{open}	same lock-in, open loop (detector only, no recurrence); $A_\Omega/A_{\mathrm{open}}$ dimensionless	mV
I, Q	in-phase, quadrature lock-in components	mV
C^*, J^*	coupling at the Hopf (criticality)	–
Laminar model (LaNMM)		
P_1, P_2	deep (alpha) and superficial (gamma) pyramidal populations	–

Symbol	**Meaning**	**Units**
$\mathrm{PV}, \mathrm{SS}, \mathrm{SST}$	parvalbumin, stellate, somatostatin interneurons	–
I_0	flat (DC) drive offset in $e(t)$	mV
Exact mean field (NMM2 / MPR) and QIF network		
$r_E,\ r_I$	population firing rates (E, I)	s^{-1}
$v_E,\ v_I$	mean membrane potentials (E, I), in dimensionless MPR units	–
$\eta_j,\ \bar{\eta}$	per-neuron excitability; mean excitability (= common drive, bifurcation param.)	–
Δ	Lorentzian half-width of η_j (excitability heterogeneity)	–
$\tau_E, \tau_I, \tau_a, \tau_g$	E, I membrane and AMPA, GABA synaptic time constants	ms
$C\,(\equiv J)$	global synaptic coupling (inter-population gain)	–
$A_{ee}, A_{ei}, A_{ie}, A_{ii}$	synaptic weight matrix $(1, 1, 1, 2)$	–
$s_E,\ s_I$	second-order synaptic activations; $\hat{K}_a, \hat{K}_g$ their operators	s^{-1}
$V_j,\ V_{\mathrm{peak}}$	QIF neuron potential; spike/reset threshold	mV
N	neurons per population (QIF; $N = 8000$)	–
$A_\Omega / \langle r_E \rangle$	Δf modulation depth (envelope-locked timing metric)	–

Table 3: Symbols, meanings, and (nominal) units. Reused glyphs are disambiguated by context; see the head note on A and C.

B TI requires detection: the Stuart–Landau normal form amplifies but cannot demodulate

The two ingredients of the mechanism—near-Hopf *amplification* and square-law *detection*—are separable, and the universal Hopf normal form makes the separation sharp. The Stuart–Landau (SL) equation [26] is

$$\dot{z} = (a + i\omega_0)z - |z|^2 z + F(t), \qquad \gamma \equiv -a > 0 \;\; \text{(stable side)}, \tag{17}$$

with the AM field $F(t) = \varepsilon[1 + m\cos\Omega t]\cos\omega_c t$. Does z acquire any power at the beat Ω (i.e. at Δf)?

Proof (carrier-harmonic ansatz). Expand z in harmonics of the carrier with slowly varying envelopes,

$$z(t) = \sum_{k\in\mathbb{Z}} Z_k(t)\, e^{ik\omega_c t}, \tag{18}$$

the Z_k varying on the slow envelope scale $\Omega \ll \omega_c$. The field forces only the $k = \pm 1$ harmonics, $F = \frac{\varepsilon}{2}(1 + m\cos\Omega t)\,(e^{i\omega_c t} + e^{-i\omega_c t})$, and the cubic is $|z|^2 z = \sum_{l,p,q} Z_l \bar{Z}_p Z_q\, e^{i(l-p+q)\omega_c t}$, feeding carrier index $k = l - p + q$. Substituting Eq. (18) into Eq. (17) and matching each $e^{ik\omega_c t}$,

$$\dot{Z}_k = [a + i(\omega_0 - k\omega_c)]\, Z_k \; - \sum_{l-p+q=k} Z_l \bar{Z}_p Z_q \; + F_k. \tag{19}$$

The *odd-k subspace is invariant and attracting*: suppose $Z_k = 0$ for every even k. Then for even k both sources vanish ($F_k = 0$, and the cubic would need $l - p + q = k$ even with l, p, q odd, which is impossible), leaving $\dot{Z}_k = [a + i(\omega_0 - k\omega_c)]Z_k$, a linear mode that decays at rate $-a = \gamma > 0$. So the solution tracks the Fourier structure of the forcing: z lives only on *odd* carrier harmonics (with sidebands at $k\omega_c \pm n\Omega$), to all orders in ε. The demodulated line at the beat is a *baseband* signal, so it would sit in the $k = 0$ component; but $Z_0 \equiv 0$, so z has identically zero weight at Δf. A square-law (*even*) nonlinearity instead produces a nonzero $k = 0$ term (from $\cos^2\omega_c t \to \frac{1}{2}$) oscillating at Ω—the demodulated line the odd cubic cannot make. Equivalently, demodulation is the $O(\varepsilon^2)$ term $\frac{1}{2}\sigma''\varepsilon^2 m$ (Eq. (10)); the SL normal form has *no* even-order term. **The SL resonator amplifies ($\sim 1/\gamma$) but does not detect.**

Numerical confirmation (Fig. S1; $a = -0.5$, $f_0 = 10$, $f_c = 100$, $\Delta f = 10$ Hz, $\varepsilon = 0.5$, $m = 1$): a square-law detector on the same field gives a Δf lock-in of $F^2|_{\Delta f} = 2.5 \times 10^{-1}$, whereas the SL state $\operatorname{Re} z$ has only $\sim 2 \times 10^{-8}$ at Δf—at the numerical floor, $\sim 10^7\times$ smaller. Driven instead *directly* in band (a tACS-like drive at Δf, no carrier), the same SL gives a forced lock-in that grows as $1/\gamma$ toward the Hopf (Fig. S1a).

The dissociation. The SL thus responds to tACS (a direct in-band drive feeds the universal near-Hopf susceptibility, $\propto 1/\gamma$) but is blind to TI (it has no square-law to demodulate the carrier). This isolates the paper's two ingredients: *amplification* is the model-independent near-Hopf susceptibility, shared with tACS and with any normal-form resonator; *detection* is the model-*specific* square-law that the SL normal form lacks. It also sharpens the founding point of §2.2: a critical resonator alone cannot demodulate—TI requires a detector.

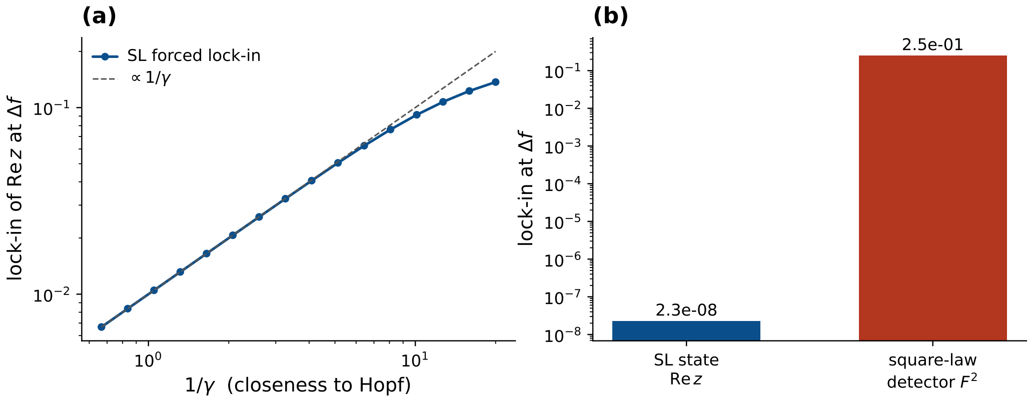

Figure S1: Stuart–Landau dissociation: amplification without detection. SL normal form (Eq. (17)). *(a)* Driven directly in band at Δf (tACS-like; weak drive, linear regime), the forced lock-in of $\mathrm{Re}\, z$ grows as $1/\gamma$ toward the Hopf—the universal near-Hopf amplification. *(b)* Under the AM *carrier* (TI), the Δf line in $\mathrm{Re}\, z$ sits at the numerical floor ($\sim 2 \times 10^{-8}$), while a square-law detector F^2 recovers it (2.5×10^{-1})—a $\sim 10^7\times$ gap. The normal form amplifies but cannot demodulate.

C Vocabulary and metrics

"Entrainment" is used in several senses, and our system can exhibit more than one, so we fix the vocabulary. We reserve *entrainment* for its dynamical-systems meaning, frequency/phase locking of a *self-sustained* oscillator to an external rhythm confined to an Arnold tongue [26], and distinguish it from two related read-outs. *(i) Forced response.* Below the Hopf the column does not oscillate on its own; the field elicits a driven response whose susceptibility we quantify by the lock-in amplitude of the read-out at the beat Δf (Eq. (14)), and there is no autonomous rhythm to entrain (J-curve, timing, tACS). *(ii) Frequency entrainment.* Above the Hopf the column free-runs at f_0; genuine entrainment means its dominant frequency is *pulled* to Δf. We test this with the dominant output frequency f_{out} (largest non-DC FFT peak) and the 1:1 criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz, mapping the Arnold tongue and locking range (Fig. S5); we use f_{out} rather than a phase-locking value [56] because, when an autonomous rhythm and a driven component coexist, instantaneous-phase estimates conflate them whereas the dominant-frequency test isolates true frequency capture. *(iii) Envelope locking / gating.* A periodic drive can instead modulate the *amplitude* of a persisting rhythm at the beat without capturing its frequency—phase–amplitude coupling [27, 28]. We quantify this by the Δf-locked component of the population rate (lock-in of r_E at Δf, equivalently the rate *folded* on the Δf beat phase—averaged over many beat cycles as a function of phase within a single cycle, so that everything not locked to Δf averages away), which isolates the driven envelope from the intrinsic rhythm. The distinction is load-bearing: the same word would otherwise denote a frequency capture (ii) and an amplitude gating (iii) that are dynamically different—so we label frequency capture *entrainment* and amplitude modulation of a persisting rhythm *gating*, reporting the metric appropriate to each (the above-Hopf QIF panel, with its $\sim$11 Hz detuning, is a case of gating, not entrainment).

Concretely for the JR entrainment analysis (Fig. S5): we place the column on the limit-cycle side—input p between the SNIC ($p \approx 114$) and the Hopf ($p \approx 315$), where it free-runs at $f_0(p)$—and drive it with $s(t) = \varepsilon \cos \Omega t$. The Arnold tongue (Fig. S5a) is the Δf lock-in over the $(\Delta f, \varepsilon)$ grid at fixed p with the 1:1-locked set outlined; the locking plateau (b) is f_{out} vs Δf (equal to Δf inside the tongue, snapping back to f_0 outside); the criticality panel (c) fixes ε and sweeps p toward the Hopf, reporting the locking range (locked-Δf width on a grid centered on $f_0(p)$) and the field-free cycle amplitude (max − min of v). Two subtleties: (i) the Δf window ($f_0 \pm 12$ Hz) is wide enough that near-Hopf locking ranges are resolved rather than grid-clipped (clipped points are flagged in Fig. S5c); (ii) as onset is approached the cycle amplitude $\to 0$ and the tongue merges into the forced-resonance bandwidth, so the supercritical locking range and the subcritical $1/\gamma$ susceptibility coincide there by construction—why responsiveness peaks at criticality on both sides.

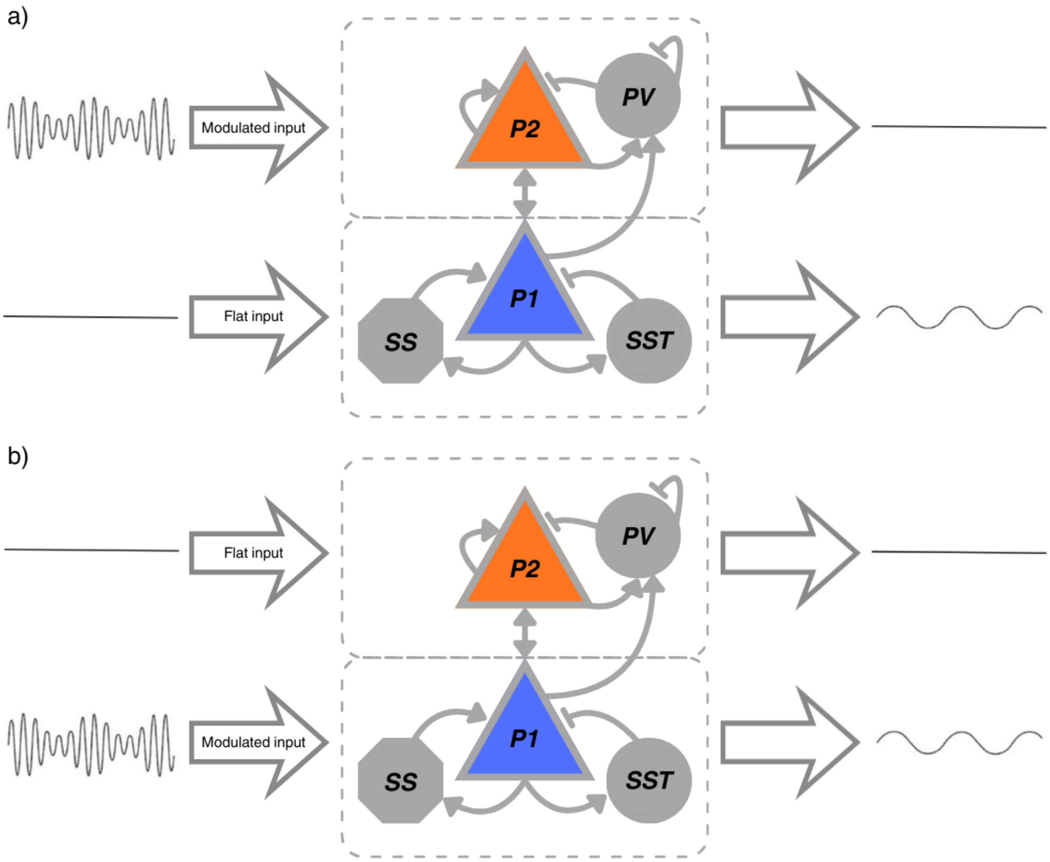

Figure S2: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the AM input drives P_2 while P_1 receives a flat drive; *(b)* the AM input drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the slow envelope is transferred to the column: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency envelope (right traces).

D Supplementary figures

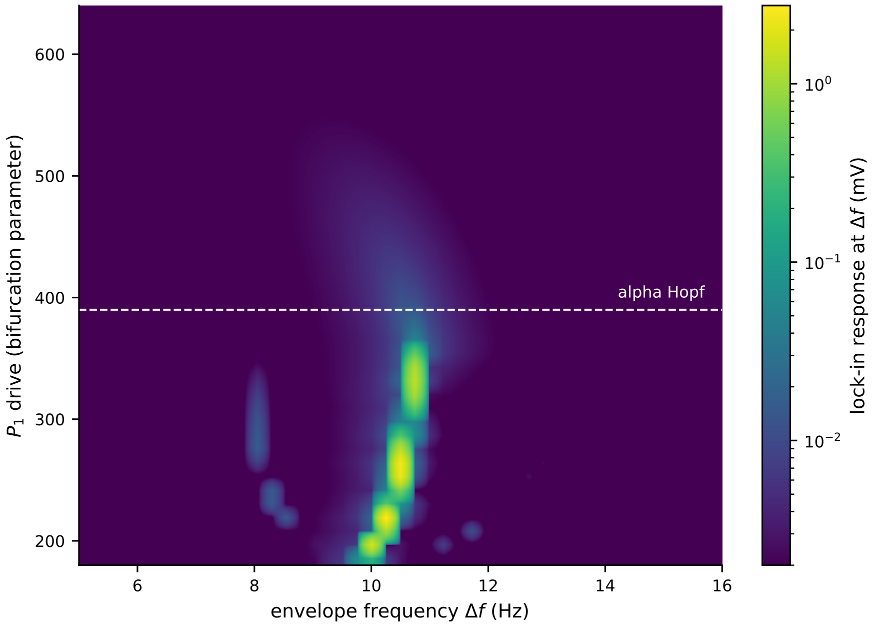

Figure S3: LaNMM alpha resonance map (NMM1, laminar). Demodulated P_1 response (log scale) over the (envelope frequency Δf, P_1 drive) plane: a ridge at the alpha natural frequency (∼10 Hz) that intensifies through the supercritical Hopf (dashed line). The analog of the JR map (Fig. S6) and the NMM2 map (Fig. S11).

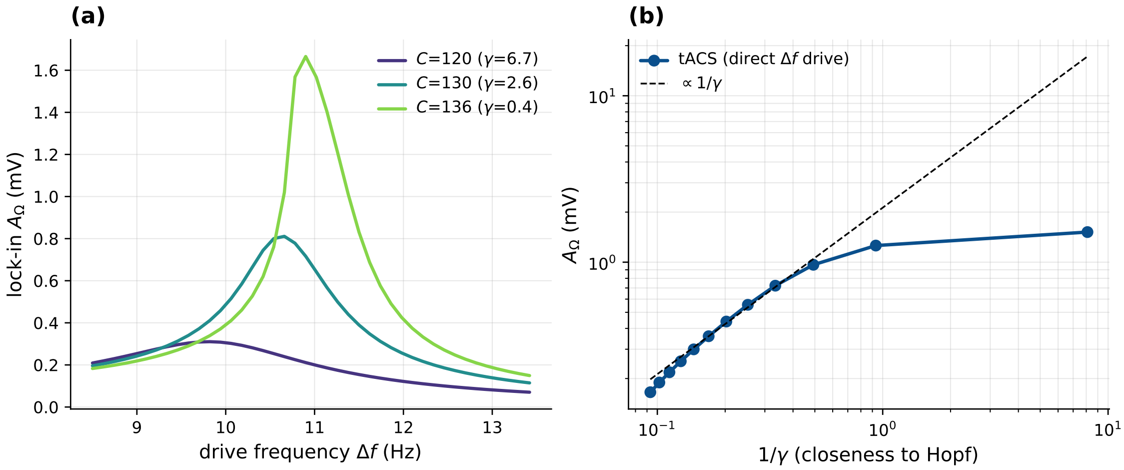

Figure S4: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the operating point pinned, sweeping the coupling $C\,(= J)$ toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 9), but linear in the field rather than square-law. Throughout, the column is a stable focus ($\gamma > 0$).

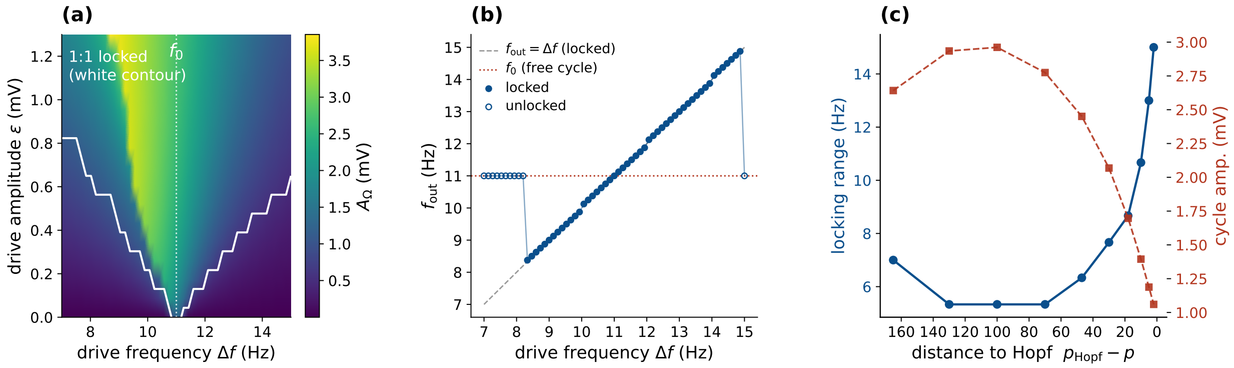

Figure S5: Above the Hopf: entrainment, not $1/\gamma$ amplification (Jansen–Rit). The column is set in its autonomous alpha limit cycle (input $p = 250$, below the input Hopf; $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid at Δf. *(a)* The 1:1 Arnold tongue—the region of $(\Delta f, \varepsilon)$ where the dominant output frequency locks to Δf—is a wedge anchored at f_0 that widens with drive amplitude. *(b)* Frequency locking at fixed drive: inside the tongue the output follows Δf (diagonal); outside, it falls back to the autonomous f_0. *(c)* Sweeping the operating point toward the Hopf at fixed drive, the locking range (tongue width) grows as the autonomous cycle amplitude $\to 0$: entrainment is easiest near criticality—the supercritical counterpart of the $1/\gamma$ forced gain (blue: locking range, left axis; red dashed: autonomous cycle amplitude, right axis). Here amplification is read as entrainment (tongue width, locking range), not as the diverging $1/\gamma$ gain of the stable-focus side (Fig. 9)—so the column is most responsive near criticality on both sides of the Hopf: to forcing below, to entrainment above.

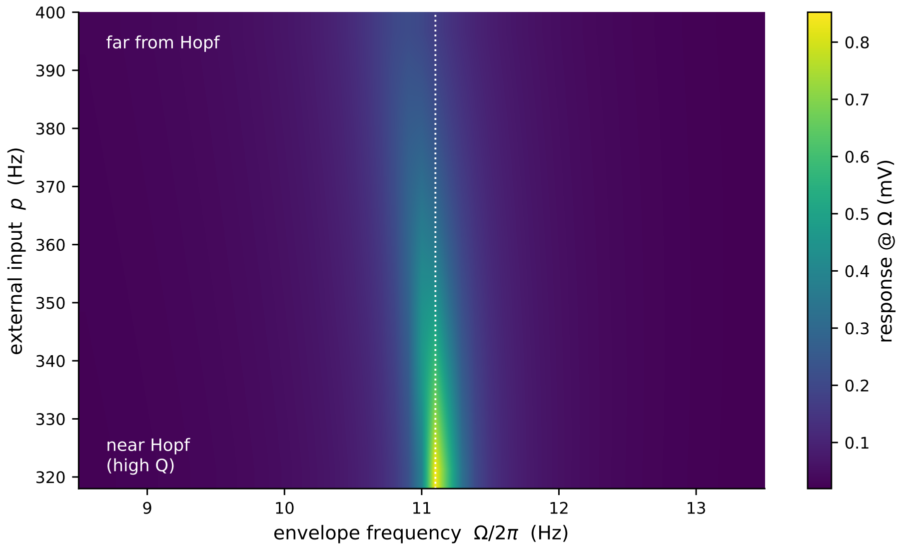

Figure S6: The resonance ridge sharpens and intensifies toward the Hopf (Jansen–Rit). Demodulated response A_Ω over the (envelope frequency, input p) plane. The ridge at $f_0 \approx 11.1$ Hz sharpens and intensifies as p approaches the Hopf from above, visualizing the $1/\gamma$ gain of Eq. (11).

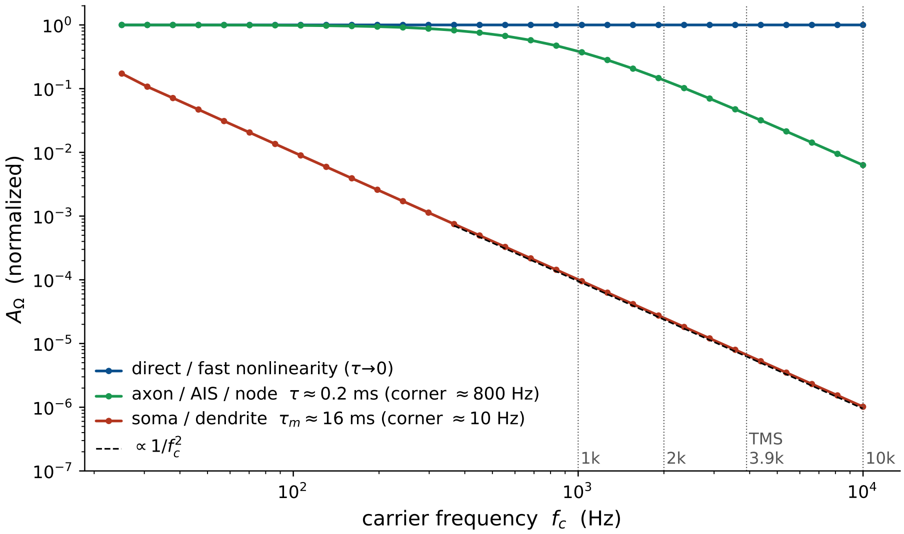

Figure S7: Somatic demodulation dies at kHz, but a fast element survives (Jansen–Rit, open-loop detector). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide), killing kHz demodulation; a fast element (axon/AIS/node, $\tau \approx 0.2$ ms) retains an appreciable response into the kHz band ($\sim$0.14 at 2 kHz, $\sim$0.04 at the TMS peak 3.9 kHz). The same membrane filter attenuates TMS at the soma, which is why neither TMS nor TI should be thought of as driving the soma at kHz.

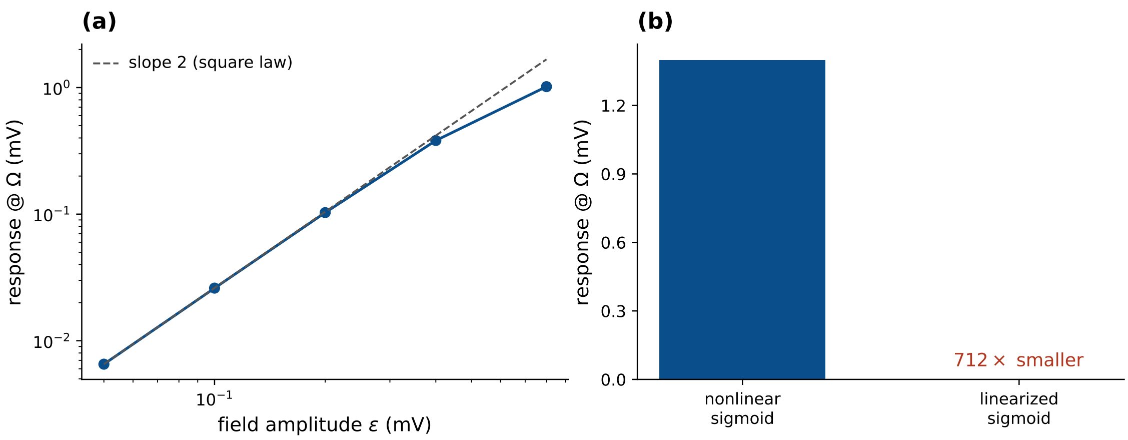

Figure S8: Verification (Jansen–Rit): square-law scaling and the linearization control. *(a)* Response vs. field amplitude; small-drive log–log slope = 1.99 (theory 2). *(b)* Linearizing the field-receiving sigmoid collapses the response 712× (1.40 → 0.0020 mV)—the nonlinearity is the demodulator.

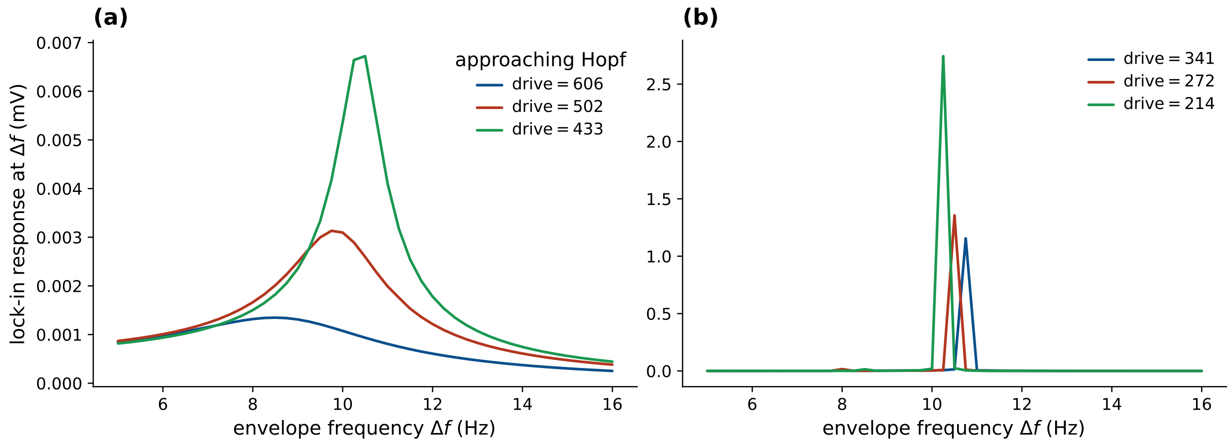

Figure S9: LaNMM alpha resonance (NMM1, laminar). Lock-in response of the deep pyramidal P_1 at the envelope frequency Δf, the P_1 (alpha) loop driven by an AM field (carrier 100 Hz) with its input as bifurcation parameter. *(a)* (stable focus, high drive) forced alpha resonance at ∼10 Hz growing as the Hopf is approached. *(b)* (limit cycle, low drive) entrainment of the autonomous alpha. Directly parallel to the JR single column (Fig. 4).

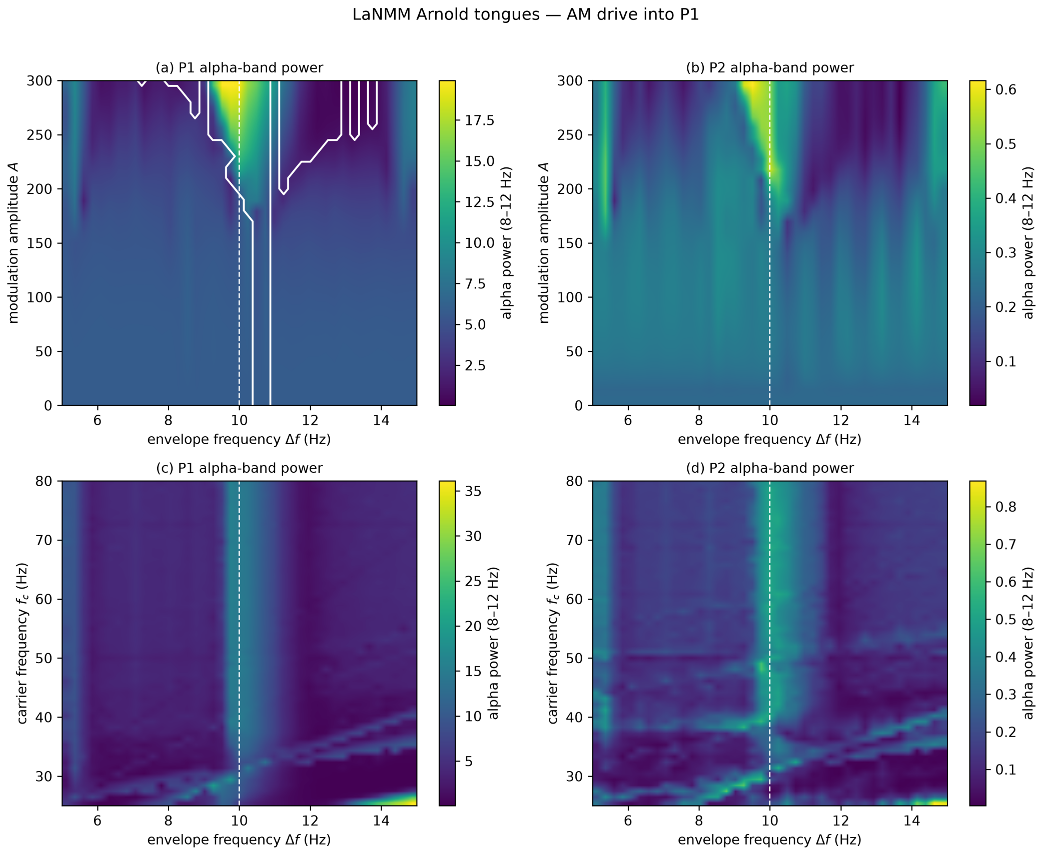

Figure S10: LaNMM Arnold tongues: AM input delivered directly to the deep P_1 (alpha) loop. *(a,b)* Alpha power of P_1 and P_2 vs. amplitude A and Δf; the white contour in *(a)* outlines the 1:1 frequency-captured set, $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix C), computed from the dominant output frequency because band power alone cannot separate capture from a forced response. *(c,d)* Alpha power vs. carrier frequency and Δf: P_1 responds across a *broad* carrier band (a vertical ridge at $\Delta f \approx 10$ Hz)—carrier-independent, the direct-coupling limit of Eq. (10).

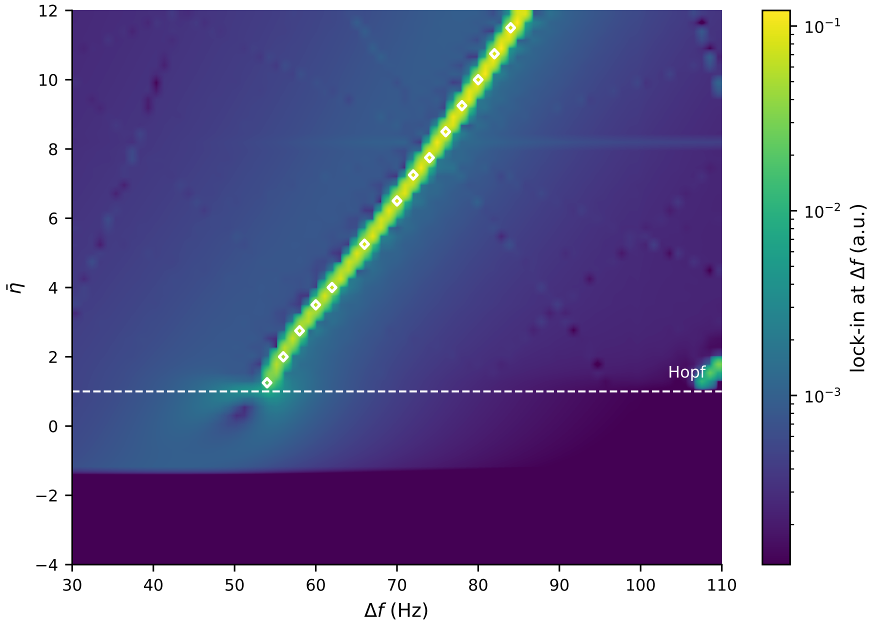

Figure S11: NMM2 PING resonance map (exact mean field). Demodulated response over the (envelope frequency Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency $f_0(\bar{\eta})$, and its diagonal slope, f_0 rising with $\bar{\eta}$, reflects the state-dependent transfer of NMM2, absent in the static-sigmoid JR and LaNMM. White markers outline the 1:1 frequency-captured set, $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix C), evaluated only above the Hopf (dashed line) where there is an autonomous rhythm to capture: entrainment tracks the ridge there, while the same color field below the Hopf is a forced response. Lock-in amplitude alone cannot separate the two, since both place power at Δf, which is why the locked set is drawn explicitly rather than inferred from the color.

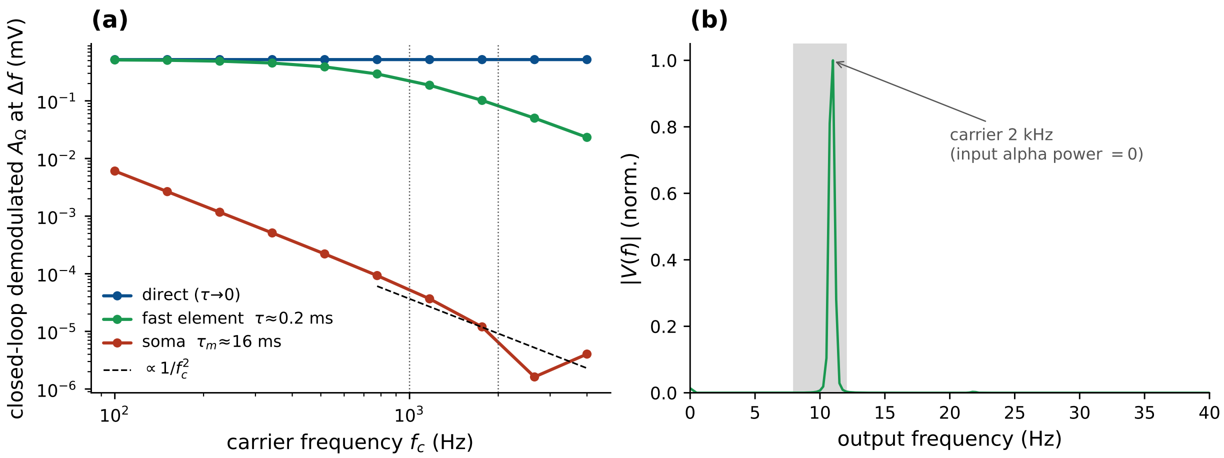

Figure S12: Demodulation at a genuine kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three coupling routes: direct ($\tau \to 0$, carrier-independent), fast element (axon/AIS/node, $\tau \approx 0.2$ ms; survives into the kHz band), and soma ($\tau_m \approx 16$ ms; collapses as $1/f_c^2$). *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: a clean alpha line at f_0, synthesized by the column although the input carries no alpha power.

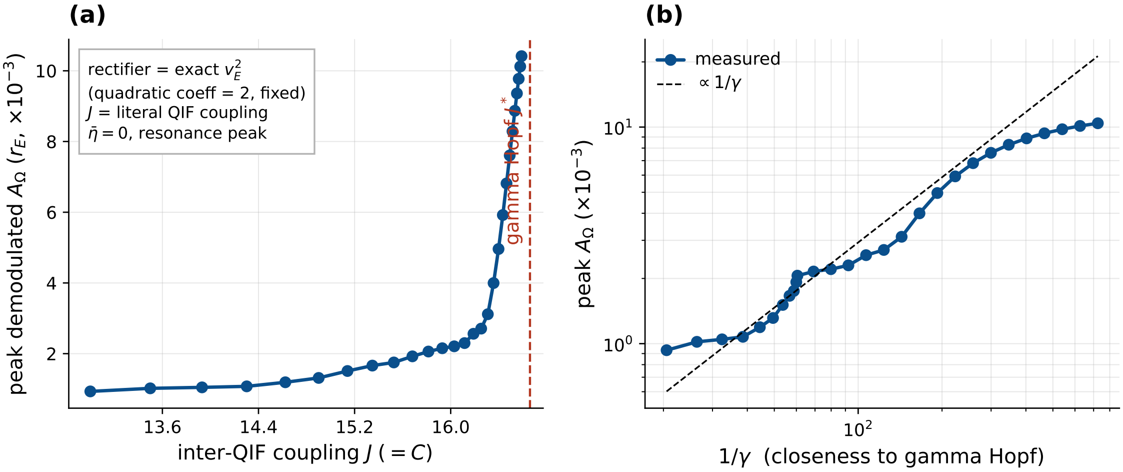

Figure S13: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). $J\,(= C)$ is the literal inter-QIF synaptic weight; the rectifier is the exact v_E^2 (fixed quadratic coefficient), so no operating-point pinning is needed. *(a)* The peak demodulated response of r_E grows as $J \to J^*$ (gamma Hopf, $\bar{\eta} = 0$; stable focus, $\gamma > 0$). *(b)* The gain grows as $1/\gamma$ (dashed), saturating near criticality—the JR law, now with coupling and rectifier both derived.

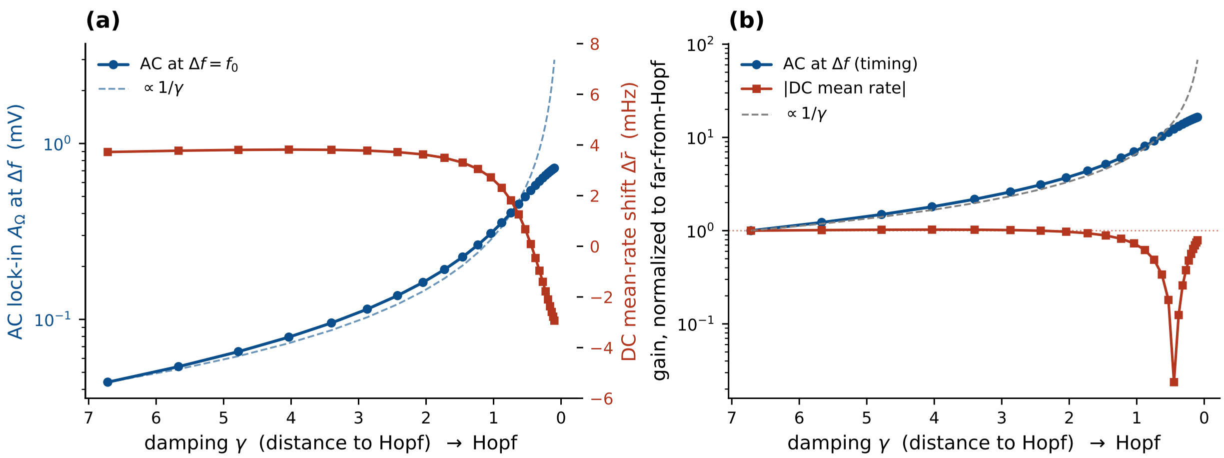

Figure S14: Timing, not rate (Jansen–Rit, detection pinned). As the column is driven toward the Hopf (from the stable side, $\gamma > 0$) through the coupling at fixed $\sigma''(v^*)$, *(a)* the AC lock-in at the resonant beat $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed), while the DC shift of the mean firing rate (red) stays flat at a few mHz. *(b)* Normalized to the far-from-Hopf value (log scale): the AC gain follows $1/\gamma$ and saturates near criticality, whereas the DC gain stays $\sim$1. The envelope locks timing, not rate—the population analog of the spike-timing-without-rate finding of [6].

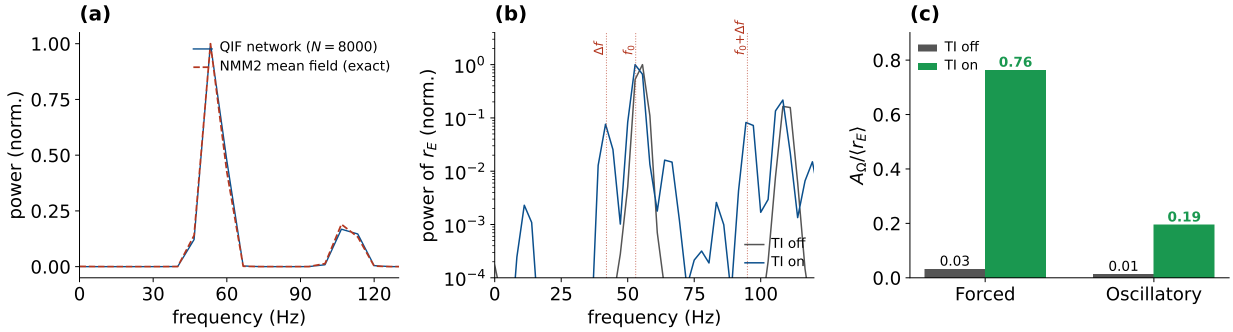

Figure S15: Timing, not rate, in the QIF network. *(a)* Power spectrum of $r_E(t)$ in the field-free (autonomous) state: the spiking network ($N = 8000$) matches the exact NMM2 mean field in the gamma rhythm (both $\sim$53 Hz) and in mean rate $\langle r_E \rangle$. *(b)* Intermodulation in the oscillatory regime (TI off vs. on, log power): with TI on the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the intrinsic $f_0 \approx 53$ Hz gamma, lighting up the direct Δf line and the *sum* sideband $f_0 + \Delta f \approx 95$ Hz, both absent when TI is off (the difference tone $f_0 - \Delta f \approx 11$ Hz sits near the floor, its amplitude set by the network transfer function rather than the mixer)—the spiking-network view of the same sigmoid frequency-mixing that drives demodulation throughout, and the sideband signature of gating rather than entrainment. *(c)* The Δf *modulation depth* $A_\Omega/\langle r_E \rangle$—the amplitude of the Δf-locked rate oscillation as a fraction of the mean rate—off vs. on: near zero with TI off (0.01–0.03), large with TI on (0.19 oscillatory, 0.76 forced), while the mean rate stays flat ($\times$1.05–1.10). We report the modulation depth rather than the off→on lock-in *ratio* because the latter divides by the near-zero field-free baseline and is realization-noisy; the depth uses two well-estimated quantities and is stable.